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Quantum Anomalies as Intrinsic Algebraic Curvature: A Unified AQFT Interpretation of Renormalization Ambiguities

Andrei T. Patrascu

FAST Foundation, Destin, FL 32541, USA; andrei.patrascu.11@ucl.ac.uk

Abstract

Quantum anomalies are traditionally understood as classical symmetries that fail to survive quantization, while experimental “anomalies” denote deviations between theoretical predictions and measured values. In this work, we develop a unified framework in which both phenomena can be interpreted through the lens of algebraic quantum field theory (AQFT). Building on the renormalization group viewed as an extension problem, we show that renormalization ambiguities correspond to nontrivial elements of Hochschild cohomology, giving rise to a deformation of the observable algebra $A * B = AB + \varepsilon \omega(A, B)$, where ω is a Hochschild 2-cocycle. We interpret ω as an intrinsic *algebraic curvature* of the net of local algebras, namely the (local) Hochschild class that measures the obstruction to trivializing infinitesimal scheme changes by inner redefinitions under locality and covariance constraints. The transported product is associative; its first-order expansion is associative up to $O(\varepsilon^2)$ while preserving the $*$ -structure and Ward identities to the first order. We prove the existence of nontrivial cocycles in the perturbative AQFT setting, derive the conditions under which the deformed product respects positivity and locality, and establish the compatibility with current conservation. The construction provides a direct algebraic bridge to standard cohomological anomalies (chiral, trace, and gravitational) and yields correlated deformations of physical amplitudes. Fixing the small deformation parameter ε from the muon $(g - 2)$ discrepancy, we propagate the framework to predictions for the electron $(g - 2)$, charged lepton EDMs, and other low-energy observables. This approach reduces reliance on ad hoc form-factor parametrizations by organizing first-order scheme-induced deformations into correlation laws among low-energy observables. We argue that interpreting quantum anomalies as manifestations of algebraic curvature opens a pathway to a unified, testable account of renormalization ambiguities and their phenomenological consequences. We emphasize that the framework does not eliminate renormalization or quantum anomalies; rather, it repackages the finite renormalization freedom of pAQFT into cohomological data and relates it functorially to standard anomaly classes.



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1. Introduction and Motivation

1.1. Quantum Anomalies Versus Experimental Discrepancies

A *quantum anomaly* is the failure of a classical symmetry of the action to be implementable at the quantum level after renormalization. Canonical examples include the axial (ABJ) anomaly in chiral currents, the trace (Weyl) anomaly of the stress tensor, and gravitational anomalies in theories of chiral matter [1–5]. By contrast, an *experimental discrepancy*

denotes a persistent deviation between a theoretically predicted quantity and its measured value (e.g., the muon anomalous magnetic moment). These notions live at different conceptual layers: the former is a cohomological obstruction in the quantization/renormalization of symmetries, while the latter is a property of phenomenological comparison.

This work develops a framework in which renormalization ambiguities in algebraic quantum field theory (AQFT) are organized by Hochschild cohomology and thereby acquire the status of an *intrinsic algebraic curvature* of the observable algebra. Within this perspective, familiar quantum anomalies are realized as particular images of cohomology classes that govern the deformation of products of observables, while certain classes of low-energy discrepancies can be probed through universal correlation laws implied by the same deformation. The guiding principle is that changes in the renormalization scheme act as (local) automorphisms on the algebra of observables; infinitesimally, they define a 2-cocycle controlling deformations of the product. The nontrivial cohomology class of this cocycle is what we call algebraic curvature.

Scope clarification.

Throughout, *it is not claimed that quantum anomalies disappear*. Standard gauge, trace/Weyl, and gravitational anomalies remain as cohomological obstructions to simultaneously enforcing locality, covariance, and Ward identities after renormalization. The present contribution is to identify a natural Hochschild-cohomological organization of renormalization ambiguities in pAQFT and to relate the resulting class $[\omega]$ to the familiar anomaly cohomology via a canonical map Φ developed below.

1.2. AQFT, Renormalization as an Extension Problem, and the Stückelberg–Petermann Group

The AQFT approach associates each causally complete spacetime region O with C^* - or von Neumann algebra $\mathcal{A}(O)$ of observables satisfying isotony, locality (Einstein causality), and covariance [6,7]. In the perturbative setting (pAQFT), interacting observables are constructed from microcausal functionals by time-ordered products T_n that satisfy causal factorization, covariance, and microlocal spectral conditions [6–9].

Renormalization is posed as an *extension problem*: distributions defined away from coincident points must be extended to the diagonals in configuration space. The freedom in this extension is finite at each order and local in the sense of scaling degree, and it is canonically organized by the Stückelberg–Petermann renormalization group SP , a group of local field redefinitions acting on the time-ordered products (or equivalently on the Bogoliubov S -matrix) [7,10,11]. Two renormalization schemes T and T' related by an element $Z \in SP$ produce the same physical S -matrix up to local counterterms fixed by renormalization conditions; the relative Z encodes the ambiguity data.

Not a “solution” of renormalization.

We stress that we do not propose an alternative to the Epstein–Glaser/Hollands–Wald renormalization program. Renormalization is still implemented as the extension of distributions to coincident configurations subject to the usual normalization conditions. Our goal is instead to make the *finite renormalization freedom* (encoded by SP) explicit in cohomological terms, thereby isolating scheme-invariant information under locality and covariance constraints.

Renormalizability versus renormalization.

We do not claim to solve renormalization as a constructive procedure. Rather, *assuming perturbative renormalizability in the Epstein–Glaser sense*, we show that the residual finite renormalization freedom—encoded by the Stückelberg–Petermann group—is governed by a local Hochschild cohomology class. This class organizes scheme dependence under locality and covariance constraints and is what we interpret as an intrinsic algebraic curvature.

1.3. From Renormalization Freedom to Hochschild Cohomology

Let $(\mathcal{A}, \cdot)$ denote the algebra of (microcausal) observables equipped with a product induced by a choice of renormalization prescription (e.g., the time-ordered product $\cdot \equiv \cdot_T$). Let $Z \in \text{SP}$ be near the identity, and we write a formal expansion

$$Z = \text{id} + \varepsilon Z_1 + O(\varepsilon^2),$$

with ε as a bookkeeping parameter. The transformed product $A *_Z B := Z(Z^{-1}A \cdot Z^{-1}B)$ is again associative. Expanding to first order in ε one obtains

$$A * B = AB + \varepsilon \omega(A, B) + O(\varepsilon^2), \quad (1)$$

where

$$\omega(A, B) := Z_1(AB) - Z_1(A)B - AZ_1(B). \quad (2)$$

Associativity of $*$ up to $O(\varepsilon^2)$ imposes the Hochschild 2-cocycle condition

$$(\delta\omega)(A, B, C) \equiv A\omega(B, C) - \omega(AB, C) + \omega(A, BC) - \omega(A, B)C = 0, \quad (3)$$

i.e., $\omega \in Z^2(\mathcal{A}, \mathcal{A})$ with respect to the Hochschild coboundary δ [12]. Conversely, a Hochschild 2-cocycle determines the first-order part of an associative deformation of the product. Coboundaries $\omega = \delta\eta$ correspond to trivial deformations induced by inner changes of variables; the cohomology class $[\omega] \in HH^2(\mathcal{A}, \mathcal{A})$ measures the obstruction to trivialization and is independent of the representative Z_1 . We refer to this class $[\omega]$ as the *algebraic curvature* of the observable algebra relative to the chosen renormalization scheme.

Two structural constraints are immediate. First, the $*$ -structure: to preserve $(A * B)^* = B^* * A^*$ at $O(\varepsilon)$, it suffices that

$$\omega(A, B)^* = \omega(B^*, A^*). \quad (4)$$

Why Hochschild cohomology appears (conceptual bridge).

The appearance of Hochschild cohomology is a standard feature of associative deformation theory: infinitesimal changes of an associative product are classified by Hochschild 2-cocycle modulo coboundaries. In pAQFT, finite renormalizations act by local redefinitions (the Stückelberg–Petermann group) on the algebra of observables; expanding a scheme transformation $Z = \text{id} + \varepsilon Z_1 + O(\varepsilon^2)$ produces the bilinear map ω in (2). Associativity of the transported product then implies the cocycle condition (3), while changes of representative $Z_1 \mapsto Z_1 + \text{ad}_\eta$ shift ω by a Hochschild coboundary. Thus, $[\omega]$ packages the scheme-change data in a cohomology class rather than introducing new degrees of freedom.

Second, locality: if A and B are localized in spacelike separated regions, microcausality demands $[A, B] = 0$. A sufficient condition for microcausality to persist to $O(\varepsilon)$ is that $\omega(A, B) - \omega(B, A) = 0$ whenever A and B are spacelike separated and that ω be *local* (supported on coincident configurations in the sense of microlocal analysis) [6–8]. We will work within the pAQFT category, where such locality conditions are natural consequences of the scaling degree bounds that govern extensions.

1.4. Interpretation as Intrinsic Algebraic Curvature

The term “curvature” is used in the following precise, algebraic sense. The family of renormalization prescriptions forms a torsor for SP; infinitesimally, changes in prescription are generated by Z_1 . The obstruction to trivializing the induced first-order deformation of the product is the Hochschild cohomology class $[\omega]$. This class is invariant under changes

of representative $Z_1 \mapsto Z_1 + \text{ad}_\eta$ and does not depend on a particular presentation. It therefore captures an intrinsic piece of the structure of the observable algebra endowed with a renormalization-compatible product. In geometric language, $[\omega]$ plays the role of a curvature 2-form for the (flat when trivial) bundle of products over the groupoid of renormalization schemes. When $[\omega] \neq 0$, there is a cohomological obstruction to globally identifying all prescriptions by inner conjugations, and this obstruction manifests in controlled deformations of composite observables.

What “curvature” is *not*

The term *curvature* is used here in a strictly algebraic and scheme-geometric sense. It does not refer to spacetime curvature, nor does it posit a new propagating field. Rather, $[\omega]$ measures the obstruction to globally identifying renormalization prescriptions by *local inner* redefinitions once locality and covariance are enforced. In this sense, it plays the role of a curvature for the “bundle of products” over the renormalization groupoid.

1.5. Relation to Known Anomaly Mechanisms

Standard quantum anomalies arise from the impossibility of simultaneously maintaining locality, covariance, and the classical Ward identities after renormalization; they therefore persist as genuine obstructions, and our framework provides an AQFT-level cohomological organization of the corresponding ambiguity data. In BRST language, anomalies are characterized by cohomology classes of local functionals obeying the Wess–Zumino consistency condition [4]. In the pAQFT setting, time-ordered products implement renormalization in configuration space, and their ambiguity is encoded by local maps acting on functionals. We will show that the first-order deviation of these maps induces a ω whose cohomology class maps functorially to the usual anomaly classes: axial (ABJ) [1,2], trace/Weyl [3], and (in appropriate chiral settings) gravitational [5]. The bridge proceeds through two steps: (i) an analysis of the Ward–Takahashi identities in the deformed product and their contact terms, and (ii) a comparison of the resulting cocycles with the BRST cohomology in the space of local functionals.

1.6. Ward–Takahashi Identities and Vertex Structures

Consider a conserved current J^μ associated with a gauge symmetry in the classical theory. Let Γ^μ denote the one-particle-irreducible (1PI) vertex function for the corresponding coupling to a Dirac field in momentum space. The Ward–Takahashi identity (WTI) reads

$$q_\mu \Gamma^\mu(p', p) = S^{-1}(p') - S^{-1}(p), \quad (5)$$

where S is the renormalized fermion propagator. In our framework, the deformed product $*$ modifies composite operators and thereby the definition of the vertex. At the first order in ε , the induced change in the vertex is linear in ω and can be organized as a sum of local contact terms plus terms proportional to standard form factors (e.g., Dirac and Pauli). A sufficient condition for (5) to hold to $O(\varepsilon)$ is that ω acts as a *biderivation* on the current algebra modulo local counterterms fixed by renormalization conditions; specifically, the map $A \mapsto \omega(A, \cdot)$ must intertwine with the derivations generated by the symmetry. This constraint is compatible with (4) and with locality, and it will be formulated and proved within pAQFT. The upshot is that identifiable pieces of ω project to the Pauli form factor $F_2(0)$ (and to CP-odd dipole operators when parity is violated), providing a foothold for phenomenological correlation tests without introducing ad hoc form factors.

1.7. Phenomenological Impetus and Scope

The formalism presented here is developed with two complementary objectives. The first is to supply a mathematically controlled interpretation of renormalization ambiguities as classes in Hochschild cohomology (algebraic curvature) in a setting that is local and covariant, namely pAQFT on globally hyperbolic spacetimes [6,7,9].

All constructions use perturbative algebraic quantum field theory (pAQFT) with Epstein–Glaser renormalization on configuration space (and the locally covariant refinement of Hollands–Wald); no alternative quantization scheme is introduced.

Second, to identify robust, *dimensionless* correlation laws among low-energy observables that are insensitive to short-distance details once a small deformation parameter ε is calibrated in one channel. We will primarily consider flavor-diagonal electroweak observables where the mapping from ω to effective operators is under perturbative control (anomalous magnetic moments and electric dipole moments of charged leptons), postponing gravitational sectors and cosmological applications to an outlook discussion where additional scales and decoupling issues arise.

1.8. Main Contributions

- We construct, within pAQFT, a deformation of the observable product $A * B = AB + \varepsilon \omega(A, B) + O(\varepsilon^2)$ induced by infinitesimal changes of renormalization prescription and prove that ω is a Hochschild 2-cocycle. The cohomology class $[\omega]$ is identified as an intrinsic algebraic curvature associated with the net of local algebras.
- We establish conditions ensuring preservation of the $*$ -structure, positivity on a dense domain, and locality to the first order in ε , and we derive a sufficient criterion for the Ward–Takahashi identities to hold up to $O(\varepsilon)$.
- We develop a functorial bridge between $[\omega]$ and standard anomaly cohomology (axial, trace, gravitational) via the structure of time-ordered products and BRST complexes, thereby unifying several anomaly mechanisms at the algebraic level.
- We formulate dimensionless correlation relations among low-energy observables that follow from the tensorial decomposition of ω , and we outline how a single calibration of ε leads to predictions for other channels with controlled uncertainties.

1.9. Organization of the Paper

Section 2 reviews the AQFT/pAQFT setting, the microlocal framework for renormalization, and the Stückelberg–Petermann group. Section 3 introduces Hochschild cohomology and proves that renormalization-induced deformations are controlled by a cocycle ω . Section 4 gives explicit constructions for ω in free theories and discusses locality and scaling. Section 5 derives the Ward–Takahashi identities in the deformed theory and analyzes the induced vertex structures. Section 6 relates $[\omega]$ to standard anomaly cohomology. Section 7 develops phenomenological correlation relations and discusses their robustness. Section 8 discusses limitations and possible extensions. Appendices A–E provide technical proofs and microlocal background.

Notation and conventions. We work in four spacetime dimensions with signature $(+, -, -, -)$. Unless stated otherwise, $\hbar = c = 1$. The formal parameter ε tracks the first-order deformation of the product induced by changes in renormalization prescription; it is distinct from any kinetic-mixing or phenomenological parameter bearing the same letter in other contexts.

For readability, the Introduction emphasizes motivation and conceptual structure; technical constructions and proofs are concentrated in later sections and appendices.

2. Preliminaries on AQFT and Renormalization Ambiguities

2.1. Local Nets, States, and Dynamics

We recall the algebraic (Haag–Kastler) framework for relativistic quantum field theory on a globally hyperbolic spacetime (M, g) (in Minkowski spacetime, this specializes to the traditional axioms; in general (M, g) , we adopt the locally covariant reformulation). To each causally convex region $O \Subset M$, one assigns a unital C^* - (or von Neumann) algebra of observables $\mathcal{A}(O)$ such that

1. **Isotony:** $O_1 \subset O_2 \Rightarrow \mathcal{A}(O_1) \subset \mathcal{A}(O_2)$.
2. **Locality (Einstein causality):** If $O_1 \perp O_2$ (spacelike separated), then $[A_1, A_2] = 0$ for all $A_i \in \mathcal{A}(O_i)$.
3. **Covariance:** There exists an action of the isometry group (Poincaré in Minkowski) by automorphisms α_g with $\alpha_g(\mathcal{A}(O)) = \mathcal{A}(gO)$.
4. **Time-slice axiom:** If O contains a Cauchy surface for $\tilde{O}$, then $\mathcal{A}(O) = \mathcal{A}(\tilde{O})$.

A *state* is a positive, normalized linear functional $\omega : \mathcal{A}(M) \rightarrow \mathbb{C}$; the GNS construction yields a representation $(\pi_\omega, \mathcal{H}_\omega, \Omega_\omega)$. In Minkowski space, physically relevant local algebras are typically of type III, whereas the perturbative constructions used below yield topological $*$ -algebras completed with respect to Hadamard two-point functions [6,7,13,14].

2.2. Microcausal Functionals and the Free-Field Star Product

Let $\mathfrak{E}(M)$ denote a suitable space of classical field configurations (e.g., smooth sections of a vector bundle). A *local functional* $F : \mathfrak{E}(M) \rightarrow \mathbb{C}$ is of the form $F(\varphi) = \int f(x, \varphi(x), \nabla\varphi(x), \dots) \, \text{dvol}_g(x)$ with an f polynomial in jets and compactly supported. Its n -th functional derivative $F^{(n)}(\varphi) \in \mathcal{D}'(M^n)$ satisfies *microcausality* if

$$\text{WF}(F^{(n)}(\varphi)) \cap (\overline{V}_+^n \cup \overline{V}_-^n) = \emptyset,$$

i.e., its wavefront set avoids unions of future/past lightcones in each cotangent factor [6,7,15]. The space of *microcausal functionals* $\mathfrak{F}_\mu(M)$ is stable under a pointwise product and functional differentiation.

Fix a Hadamard two-point function ω_2 for the free field; define the bidifferential operator

$$\Gamma_{\omega_2} := \frac{1}{2} \int_{M \times M} \omega_2(x, y) \frac{\delta}{\delta\varphi(x)} \otimes \frac{\delta}{\delta\varphi(y)} \, \text{dvol}_g(x) \, \text{dvol}_g(y).$$

Then, the (normal ordered) *free-field star product* on $\mathfrak{F}_\mu(M)$ is given by

$$F \star G := \mu \circ \exp(\hbar \Gamma_{\omega_2})(F \otimes G) = \sum_{n=0}^{\infty} \frac{\hbar^n}{n!} \langle \omega_2^{\otimes n}, F^{(n)} \otimes G^{(n)} \rangle, \quad (6)$$

where μ is the pointwise multiplication and the brackets denote the distributional pairing (well-defined by Hörmander's criterion and microcausality). The resulting topological $*$ -algebra $(\mathfrak{A}_{\omega_2}(M), \star)$ depends on the choice of ω_2 by an inner (coherent-state) equivalence; observables are thus defined modulo this equivalence [6,7].

2.3. Time-Ordered Products and Causal Factorization

Interacting observables are constructed from *time-ordered products*

$$T_n : \underbrace{\mathfrak{F}_{\text{loc}}(M) \times \cdots \times \mathfrak{F}_{\text{loc}}(M)}_n \rightarrow \mathfrak{A}(M), \quad n \geq 0,$$

satisfying axioms of symmetry, causal factorization, unit, microlocal spectrum condition, covariance, scaling degree bounds, and field independence [6–9,16]. Causal factorization reads as follows: if $\text{supp } F_i < \text{supp } F_j$ for all $i \in I, j \in J$ (i.e., all points in $\text{supp } F_i$ are not later than any in $\text{supp } F_j$), then

$$T_{|I|+|J|}((F_i)_{i \in I}, (F_j)_{j \in J}) = T_{|I|}((F_i)_{i \in I}) \star T_{|J|}((F_j)_{j \in J}).$$

For free fields, a compact formula is

$$T = \exp(\hbar \Gamma_{H_F}), \quad \Gamma_{H_F} := \frac{1}{2} \int H_F(x, y) \frac{\delta}{\delta \varphi(x)} \otimes \frac{\delta}{\delta \varphi(y)} \text{dvol}_g(x) \text{dvol}_g(y),$$

where H_F is the Feynman propagator (Hadamard parametrix with $i0$ -prescription). Interactions are introduced via the Bogoliubov S -matrix

$$S(V) := T \exp\left(\frac{i}{\hbar} V\right), \quad V \in \mathfrak{F}_{\text{loc}}(M),$$

and the relative S -matrix

$$S_V(F) := S(V)^{-1} \star S(V + F).$$

The interacting (retarded) field is obtained by functional differentiation at zero coupling:

$$R_V(F) := \left. \frac{\text{d}}{\text{d}\lambda} \right|_{\lambda=0} S_V(\lambda F).$$

Causal factorization of T ensures the *Bogoliubov causality* of R_V , while the time-slice axiom holds provided T is constructed covariantly and locally [6,9,14].

2.4. Microlocal Extension and the Epstein–Glaser Program

The inductive construction of T_n (Epstein–Glaser) reduces to extending distributions defined on configuration space with *thin* diagonals removed to distributions supported also on the diagonals, under control of scaling degrees. Here, configuration space refers to M^n for n insertions; “coincident points” correspond to the thin (partial) diagonals where one or more insertion points coincide, i.e., loci of the form $x_i = x_j$. The extension ambiguity is supported on these diagonals and is controlled by the scaling degree. Let $t \in \mathcal{D}'(\mathbb{R}^{Nd} \setminus \{0\})$; its (Steinmann) *scaling degree* at the origin is

$$\text{sd}_0(t) := \inf \left\{ \delta \in \mathbb{R} : \lim_{\lambda \downarrow 0} \lambda^\delta t(\lambda x) = 0 \text{ in } \mathcal{D}'(\mathbb{R}^{Nd} \setminus \{0\}) \right\}.$$

Theorem 1 (Extension with minimal scaling degree [8,17]). *If $t \in \mathcal{D}'(\mathbb{R}^{Nd} \setminus \{0\})$ has a finite scaling degree $\text{sd}_0(t) < \infty$, then there exists an extension $\bar{t} \in \mathcal{D}'(\mathbb{R}^{Nd})$ with $\text{sd}_0(\bar{t}) = \text{sd}_0(t)$. If $\omega := \text{sd}_0(t) - Nd \geq 0$, the extension is not unique; the ambiguity space is finite-dimensional and spanned by derivatives $\partial^\alpha \delta$ with $|\alpha| \leq \omega$.*

In the EG recursion for T_n , the singular distributions arise from concatenations of Feynman propagators on configuration space. The theorem implies the following:

- *Existence:* T_n exists for all n with the stated axioms, by induction on n .
- *Locality and covariance:* If the choice of extension is local and covariant (in the sense of natural transformations under embeddings of spacetimes), then so is T_n [6,8,14,18].
- *Finite renormalization freedom:* At each order, ambiguities correspond to local counterterms supported on diagonals, with the dimension bounded by the degree of divergence ω .

Thus, renormalization is precisely the *freedom of extension* at coinciding points, controlled by scaling degree and microlocal spectral properties (Hadamard condition) [6,19].

2.5. Normalization Conditions and Renormalization Schemes

To fix T_n uniquely, one imposes normalization conditions (in the spirit of EG/Scharf):

- N0. *Symmetry*: T_n is symmetric in its arguments.
- N1. *Causal factorization* (as above).
- N2. *Unit*: $T_0 = 1$ and $T_1(F) = F$ for linear F .
- N3. *Covariance*: T_n is natural with respect to isometries/embeddings.
- N4. *Microlocal spectrum*: wavefront set bounds are consistent with Hadamard structure.
- N5. *Scaling*: scaling degree bounds are compatible with canonical dimensions.
- N6. *Unitarity*: $T_n^\dagger = \bar{T}_n$ (causal unitarity).
- N7. *Field independence*: functional derivatives commute appropriately with T_n .

Different choices of extensions consistent with N0–N7 define distinct *renormalization schemes*. The space of schemes is an affine space modeled on local counterterms (polynomial in fields and derivatives) graded by canonical dimension or scaling degree.

2.6. The Stückelberg–Petermann Renormalization Group

Let $\mathcal{F}_{\text{loc}}$ be the space of local functionals. The *Stückelberg–Petermann group* SP is the group of formal, local redefinitions $Z : \mathcal{F}_{\text{loc}} \rightarrow \mathcal{F}_{\text{loc}}$ of the form

$$Z = \text{id} + \sum_{k \geq 1} \hbar^k Z_k, \quad Z(F + G) = Z(F) + Z(G) + \text{local terms vanishing at } F = G = 0,$$

compatible with support, covariance, and degree (it acts as a formal diffeomorphism on the space of interactions). The *Main Theorem of Renormalization* in the causal framework can be stated as follows:

Theorem 2 (Main theorem, causal version [10,11]). *Let T and T' be two time-ordered product families satisfying the axioms N0–N7. Then there exists a unique $Z \in \text{SP}$ such that the corresponding Bogoliubov S -matrices obey*

$$S'(V) = S(Z(V)), \quad \forall V \in \mathcal{F}_{\text{loc}}.$$

Moreover, the composition of renormalization schemes corresponds to the composition in SP.

In particular, the physical content encoded by relative S -matrices is invariant under Z , while individual composite operators transform covariantly. The difference between schemes appears as local reparametrizations of couplings and fields, and as finite redefinitions of composite operators. This group-theoretic structure will be the starting point for our algebraic deformation analysis.

2.7. Retarded Products and the Algebra of Interacting Observables

Define the multilinear *retarded products* $R_{n,1}$ by

$$R_{n,1}(F_1, \dots, F_n; G) := \frac{\partial^n}{\partial \lambda_1 \dots \partial \lambda_n} \Big|_0 S \left(\sum_{i=1}^n \lambda_i F_i \right)^{-1} \star S \left(\sum_{i=1}^n \lambda_i F_i + G \right).$$

They satisfy causal support properties: $\text{supp } R_{n,1}(\vec{F}; G) \subset \bigcap_j J^+(\text{supp } F_j) \cap J^-(\text{supp } G)$ and obey combinatorial identities mirroring BPHZ forests but formulated in configuration space [6,16]. The interacting algebra generated by $R_V(F)$ (for a fixed interaction V)

inherits the net structure and the time-slice property from T_n , and its dependence on the renormalization scheme is governed by SP as above.

2.8. Example: φ^4 in Four Dimensions

For concreteness, consider the scalar model with $V = \int (\lambda/4!) \varphi^4 \text{dvol}_g$. Power counting gives a degree of divergence $\omega(\Gamma) = 4L - 2I$ for a graph with loops L and internal lines I . In EG language, the singular part at order n is supported on the union of partial diagonals in M^n ; scaling degree bounds restrict the ambiguity to local counterterms of the schematic form

$$\delta \mathcal{L} = \delta Z_\varphi (\partial \varphi)^2 + \delta m^2 \varphi^2 + \delta \lambda \varphi^4 + \dots,$$

with coefficients polynomial in λ and distributions built from curvature if (M, g) is curved (local covariance). Changing the scheme amounts to acting by some $Z \in \text{SP}$ that reparametrizes $(\lambda, m^2, Z_\varphi)$ and modifies composite operators (e.g., Wick powers) by local terms, consistent with Hollands–Wald’s classification of Wick polynomials [8,9].

2.9. Summary of Structural Consequences

The EG/pAQFT framework yields the following:

- A functorial assignment $O \mapsto \mathcal{A}(O)$ with time-ordered products T_n encoding interactions and retarded products $R_{n,1}$ encoding dynamics.
- A finite-dimensional space of renormalization ambiguities at each order, localized on diagonals and controlled by scaling degree.
- A global, group-theoretic organization of scheme changes via SP, acting by formal local redefinitions on interactions and composite operators.

In Section 3, we will use these facts to identify the *first-order* effect of a scheme change with a Hochschild 2-cocycle ω that controls an associative deformation of the observable product. The associated cohomology class will be our *algebraic curvature*.

3. Algebraic Curvature as Hochschild Cohomology

3.1. Hochschild Cochains, Differential, and Gerstenhaber Bracket

Let $\mathcal{A}$ be a (topological) associative $*$ -algebra over $\mathbb{C}$ with multiplication $m : \mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}$, $m(A, B) = AB$. The space of n -cochains $C^n(\mathcal{A}, \mathcal{A})$ consists of continuous n -linear maps $\phi : \mathcal{A}^{\otimes n} \rightarrow \mathcal{A}$ (with $C^0(\mathcal{A}, \mathcal{A}) \cong \mathcal{A}$). The *Hochschild differential* $\delta : C^n \rightarrow C^{n+1}$ is

$$\begin{aligned} (\delta \phi)(A_1, \dots, A_{n+1}) &= A_1 \phi(A_2, \dots, A_{n+1}) \\ &+ \sum_{i=1}^n (-1)^i \phi(A_1, \dots, A_i A_{i+1}, \dots, A_{n+1}) + (-1)^{n+1} \phi(A_1, \dots, A_n) A_{n+1}. \end{aligned} \quad (7)$$

One verifies $\delta^2 = 0$; the Hochschild cohomology $HH^\bullet(\mathcal{A}, \mathcal{A})$ is $Z^\bullet/B^\bullet$ with $Z^\bullet = \ker \delta$ and $B^\bullet = \text{im } \delta$ [20,21].

The space $C^\bullet(\mathcal{A}, \mathcal{A})[1]$ (degree shifted by +1) carries the *Gerstenhaber bracket* $[\cdot, \cdot]_G$ that turns it into a graded Lie algebra [12,22]. For $\phi \in C^p$, $\psi \in C^q$,

$$[\phi, \psi]_G = \phi \circ \psi - (-1)^{(p-1)(q-1)} \psi \circ \phi, \quad (\phi \circ \psi)(A_1, \dots, A_{p+q-1}) := \sum_{i=1}^p (-1)^{(i-1)(q-1)} \phi(\dots, \psi(A_i, \dots), \dots). \quad (8)$$

The associative product m is a 2-cochain obeying $[m, m]_G = 0$. The Hochschild differential is $\delta \phi = [m, \phi]_G$, so $\delta^2 = [m, [m, \cdot]] = \frac{1}{2} [[m, m], \cdot] = 0$ by graded Jacobi.

3.2. First-Order Deformations and Classification by HH^2

A formal associative deformation of m is a $\mathbb{C}[[\varepsilon]]$ -bilinear product

$$A * B = m(A, B) + \varepsilon \omega(A, B) + \varepsilon^2 \omega_2(A, B) + \cdots, \quad (9)$$

with $\omega_k \in C^2(\mathcal{A}, \mathcal{A})$, such that $(A * B) * C = A * (B * C)$ in $\mathbb{C}[[\varepsilon]] \otimes \mathcal{A}$. Expanding the associativity constraint order-by-order yields the *Maurer–Cartan tower*

$$O(\varepsilon): [m, \omega]_G = 0 \iff \delta\omega = 0, \quad (10)$$

$$O(\varepsilon^2): [m, \omega_2]_G = -\frac{1}{2}[\omega, \omega]_G, \quad \text{etc.} \quad (11)$$

Thus, a necessary and sufficient condition for a *first-order* associative deformation is that ω be a Hochschild 2-cocycle. Two first-order deformations ω and ω' are *equivalent* if there exists a formal automorphism $\Phi = \text{id} + \varepsilon \eta + \cdots$ with $\eta \in C^1(\mathcal{A}, \mathcal{A})$ such that

$$\omega' = \omega + \delta\eta. \quad (12)$$

Hence, equivalence classes of first-order associative deformations are classified by $HH^2(\mathcal{A}, \mathcal{A})$ [12,23,24]. In geometric language, the moduli of associative structures modulo inner equivalence has tangent space HH^2 at m .

3.3. Scheme Changes $\Rightarrow$ Hochschild 2-Cocycles

Let T be a fixed renormalization prescription (family of time-ordered products) in pAQFT and let SP be the Stückelberg–Petermann group of local, formal redefinitions acting on interactions and composites. For $Z \in \text{SP}$ close to the identity, define a new product on $\mathcal{A}$ by transport of structure

$$A *_Z B := Z(Z^{-1}A \cdot Z^{-1}B), \quad (13)$$

where $\cdot$ denotes the original product (time-ordered or otherwise) associated with T . Writing $Z = \text{id} + \varepsilon Z_1 + O(\varepsilon^2)$, a direct calculation gives at first order

$$A * B = AB + \varepsilon \omega(A, B) + O(\varepsilon^2), \quad \omega(A, B) := Z_1(AB) - Z_1(A)B - AZ_1(B). \quad (14)$$

Proposition 1 (Cocycle property of ω). *The bilinear map ω in (14) obeys $\delta\omega = 0$. Moreover, if $Z_1 \mapsto Z_1 + \text{ad}_\eta$ with $\eta \in C^1(\mathcal{A}, \mathcal{A})$, then $\omega \mapsto \omega + \delta\eta$. Hence, the cohomology class $[\omega] \in HH^2(\mathcal{A}, \mathcal{A})$ is well-defined by the scheme-change orbit of T .*

Sketch. Associativity of $*_Z$ implies $[m + \varepsilon\omega, m + \varepsilon\omega]_G = 0$ at $O(\varepsilon)$, i.e., $[m, \omega]_G = 0$, which is $\delta\omega = 0$ by the identity $\delta = [m, \cdot]$. Under $Z_1 \mapsto Z_1 + \eta \circ (\cdot)$ with η a derivation-like 1-cochain, (14) shifts by $\delta\eta$ exactly as in the standard deformation equivalence. $\square$

We call $[\omega]$ the *algebraic curvature* associated with the renormalization class of T . It is intrinsic up to inner equivalences generated by 1-cochains and, crucially, is *local* in the sense explained below.

3.4. Locality, $*$ -Structure, and Microcausality at First Order

In AQFT/pAQFT, one requires that composite operations be local and respect the causal structure. Let $\mathcal{A} = \mathcal{A}(M)$ be generated by microcausal functionals; locality of ω means that $\omega(A, B)$ depends on the jets of A, B at coincident points and is supported on the diagonals of configuration space (as a distribution-valued bidifferential operator).

Proposition 2 (Locality and microcausality). *Suppose ω arises from a renormalization scheme change Z that preserves the microlocal spectrum condition and naturality under embeddings (locally covariant). Then ω is supported on the diagonals and is a local bidifferential operator of finite order (bounded by scaling degree). In particular, if A and B are localized in spacelike separated regions, then $[A, B] = 0$ implies $[A, B]_* := A * B - B * A = O(\varepsilon^2)$.*

Sketch. In the Epstein–Glaser construction, ambiguities at order n are linear combinations of derivatives of delta distributions supported on partial diagonals, multiplied by local polynomial functionals with bounded scaling degree [8,17,25]. The induced first-order change Z_1 is local and respects causal factorization; therefore, ω computed from (14) inherits locality and finite order. Spacelike commutativity persists because the support properties forbid nontrivial contributions when arguments are causally disjoint to first order. $\square$

The $*$ -structure imposes the compatibility condition

$$(A * B)^\dagger = B^\dagger * A^\dagger \iff \omega(A, B)^\dagger = \omega(B^\dagger, A^\dagger). \quad (15)$$

This holds provided Z_1 intertwines the adjoint (causal unitarity) [9,16].

Remark 1 (Positivity to first order). *Let φ be a positive, continuous state on $\mathcal{A}$. If ω satisfies (15) and is bounded on the relevant dense domain, then for sufficiently small ε one has $\varphi(A^\dagger * A) \geq -C \varepsilon \|A\|^2$ on that domain, i.e., quasi-positivity to first order. In our perturbative context, observables are formal power series and positivity is interpreted order-by-order [7].*

3.5. Explicit Local Form of ω in pAQFT

Let T and T' be two time-ordered products obeying the pAQFT axioms. Their difference at order n is supported on the union of partial diagonals in M^n and has the general form

$$T'_n(F_1, \dots, F_n) - T_n(F_1, \dots, F_n) = \sum_{|\alpha| \leq N} \int_M c_\alpha(x) \partial^\alpha \delta(x_1 - \dots - x_n) \prod_{j=1}^n F_j^{(k_j)}(x) \, \text{dvol}_g(x), \quad (16)$$

with local coefficients c_α constrained by scaling. Specializing to $n = 2$ and transporting via (14), one obtains the typical structure

$$\omega(F, G) = \sum_{|\alpha| \leq N} \int_M C_\alpha(x) \langle \partial^\alpha \delta(x - y), F^{(1)}(x) \otimes G^{(1)}(y) \rangle \, \text{dvol}_g(x) + \text{higher functional-derivative terms}, \quad (17)$$

where C_α are locally and covariantly constructed from the metric and coupling constants (and curvature if (M, g) is curved). Equation (17) makes explicit that ω is local and supported on coincident configurations. The “higher functional-derivative terms” appear when Z_1 involves field redefinitions of order ≥ 2 in the fields.

3.6. Biderivations and Ward Identities at $O(\varepsilon)$

For later use (vertex analysis), it is convenient to single out a class of ω that act as *biderivations* modulo local contact terms:

$$\omega(A, BC) = \omega(A, B) C + B \omega(A, C) + \text{local contact terms}, \quad \omega(AB, C) = A \omega(B, C) + \omega(A, C) B + \text{l.c.t.} \quad (18)$$

Such a property holds if Z_1 is (to first order) a derivation with respect to the product modulo local counterterms fixed by normalization conditions (N0–N7). In pAQFT this amounts to the field independence and covariance of T and the naturality of Z [7,11]. When (18) is satisfied, the deformation of the Noether current algebra respects the Ward–Takahashi identities to first order after appropriate redefinition of local counterterms (Section 5).

3.7. Comparison with Standard Deformation Quantizations

It is instructive to contrast the present deformation with classical star products from deformation quantization. If (M, π) is a Poisson manifold, a formal star product is

$$f \star_{\hbar} g = fg + \frac{i\hbar}{2} \{f, g\} + \sum_{k \geq 2} \hbar^k B_k(f, g), \quad (19)$$

with bidifferential operators B_k and $B_1(f, g) - B_1(g, f) = i\{f, g\}$ [26–28]. In that setting, HH^2 detects infinitesimal classes of associative deformations compatible with the Poisson bracket and is related to Poisson cohomology. In our AQFT/pAQFT context, the algebra $\mathcal{A}$ is noncommutative from the outset; the deformation parameter ε tracks *renormalization-scheme* changes rather than quantization, and the corresponding ω is *local* (supported on diagonals).

On the C^* -algebraic side, Rieffel deformation by an $\mathbb{R}^d$ -action produces associative star products that notoriously *preserve* the C^* -norm and positivity [29]. Net-level implementations (warped convolutions) yield wedge-local quantum fields [30,31]. Our ω can be viewed as the formal, local, configuration-space analogue driven by the renormalization-group torsor rather than by an external group action.

3.8. Summary of the Cohomological Picture

The discussion above establishes the following chain:

$$\text{renormalization scheme change } Z \in \text{SP} \implies \text{first-order product deformation } \omega \implies [\omega] \in HH^2(\mathcal{A}, \mathcal{A}),$$

with $[\omega]$ invariant under inner equivalence and local in the pAQFT sense. We interpret $[\omega]$ as an *intrinsic algebraic curvature*. In the next sections we exploit this structure to control Ward identities and to relate $[\omega]$ to standard anomaly cohomology, then to extract universal correlation laws among low-energy observables.

4. Explicit Constructions of the Cocycle ω

In this section, we construct the first-order deformation cocycle ω explicitly in perturbative AQFT. We begin by displaying ω as the differential of an infinitesimal Stückelberg–Petermann map acting on local functionals and then work out concrete formulas for free scalar and Dirac fields. We emphasize locality (support on diagonals), covariance, scaling-degree bounds, and the compatibility with the $*$ -structure. Finally, we indicate conditions for nontriviality of $[\omega] \in HH^2(\mathcal{A}, \mathcal{A})$ within natural subcategories of local, covariant cochains.

4.1. Renormalization Maps as Local Differential Operators

Let $\mathcal{F}_{\text{loc}}(M)$ denote the space of local functionals (polynomial in jets, compactly supported). A renormalization prescription T (family of time-ordered products) induces a product $A \cdot B$ on an algebra $\mathcal{A}$ generated by microcausal functionals. A change of prescription $T \mapsto T'$ is encoded by $Z \in \text{SP}$ (Main Theorem), i.e., a formal, *local and covariant* redefinition $Z : \mathcal{F}_{\text{loc}} \rightarrow \mathcal{F}_{\text{loc}}$ acting functorially under isometric embeddings [7,11]. We write

$$Z = \text{id} + \varepsilon Z_1 + O(\varepsilon^2), \quad Z_1 = \sum_{r=0}^R \int_M \sum_{|\alpha| \leq N_r} C_\alpha^{(r)}(x) \mathcal{D}_\alpha^{(r)} \text{dvol}_g(x), \quad (20)$$

where $\mathcal{D}_\alpha^{(r)}$ values are *local* differential operators of total jet-order $\leq r$ acting on the integrands of local functionals, and the coefficients $C_\alpha^{(r)}$ are locally and covariantly constructed from the background (metric, couplings, curvature) with definite scaling dimension; R and

N_r are bounded by the scaling-degree constraints of the Epstein–Glaser extension at the order considered [8,9,25].

Transport of structure (Section 3) yields the deformed product

$$A * B = AB + \varepsilon \omega(A, B) + O(\varepsilon^2), \quad \omega(A, B) := Z_1(AB) - Z_1(A)B - AZ_1(B), \quad (21)$$

with $\delta\omega = 0$ and $[\omega]$ independent of the representative Z_1 modulo 1-cochains. Since Z_1 is local and covariant, ω is a *local bidifferential operator* supported on coincident configurations (partial diagonals) and bounded in order by the same scaling-degree estimates.

4.2. Scalar Fields: Wick Monomials and Time-Ordered Products

Consider a real free Klein–Gordon field on a globally hyperbolic spacetime (M, g) , with Hadamard two-point function ω_2 and Feynman parametrix H_F . For compactly supported test functions f, g , define Wick monomials (with respect to ω_2)

$$:\phi^k:(f) := \int_M :\phi(x)^k: f(x) \, \text{dvol}_g(x),$$

where the normal ordering is given by the free-field star product [6,7]. The time-ordered product of local fields is built from H_F and extended to partial diagonals by the Epstein–Glaser recursion subject to the local covariance and scaling-degree constraints [8,9].

A worked example: $:\phi^2: \times :\phi^2:$

On $M = \mathbb{R}^{1,3}$ (for concreteness), Wick’s theorem gives the point-split form

$$T(:\phi^2:(x), :\phi^2:(y)) = :\phi^2(x)\phi^2(y): + 4H_F(x, y) :\phi(x)\phi(y): + 2H_F(x, y)^2. \quad (22)$$

The last term is a distribution singular as $x \rightarrow y$ with scaling degree 4; its extension to the diagonal is *not unique*. Minimal scaling-degree extension is unique up to local counterterms supported on $x = y$:

$$H_F(x, y)^2 \rightsquigarrow (H_F^2)_{\text{ext}}(x, y) + \sum_{|\alpha| \leq 2} a_\alpha \partial^\alpha \delta(x - y), \quad (23)$$

with constants a_α (in curved backgrounds, the a_α values are replaced by local curvature polynomials). Changing the extension by Δa_α produces a shift in T_2 by a *local* term:

$$\Delta T_2(:\phi^2:(f), :\phi^2:(g)) = 2 \sum_{|\alpha| \leq 2} \Delta a_\alpha \int_M \partial^\alpha (fg)(x) \, \mathbf{1} \, \text{dvol}_g(x), \quad (24)$$

where $\mathbf{1}$ denotes the unit. Interpreting this change via (21), we obtain the corresponding contribution to ω :

$$\omega(:\phi^2:(f), :\phi^2:(g)) = 2 \sum_{|\alpha| \leq 2} \Delta a_\alpha \int_M \partial^\alpha (fg)(x) \, \mathbf{1} \, \text{dvol}_g(x). \quad (25)$$

Thus, ω is explicitly a local bidifferential operator of finite order, supported on the diagonal, with coefficients fixed by the (finite) extension freedom. By construction, $\delta\omega = 0$, and the class $[\omega]$ is independent of the presentation of Z_1 .

Further scalar examples.

For mixed monomials such as $:\phi^3: \times :\phi:$, one obtains

$$T(:\phi^3:(x), :\phi:(y)) = :\phi^3(x)\phi(y): + 3H_F(x, y) :\phi^2(x): + \text{less singular}, \quad (26)$$

and the extension ambiguity resides in the term $H_F(x, y)\delta$ when $x \rightarrow y$. A change of extension yields

$$\omega(\phi^3 : (f), : \phi : (g)) = 3 \sum_{|\alpha| \leq 1} \Delta b_\alpha \int_M \partial^\alpha (f g)(x) : \phi^2(x) : \mathrm{dvol}_g(x), \quad (27)$$

again manifestly local. Analogous formulas hold for higher-degree monomials, with the highest number of derivatives determined by the scaling degree of the singular kernel entering the unextended product; the coefficients Δb_α depend polynomially on couplings and (in curved backgrounds) on curvature scalars [8,9].

Proposition 3 (Locality, covariance, and order bounds). *For scalar Wick monomials, the cocycle ω determined by any change of extension in the Epstein–Glaser construction has the form of a finite sum of terms like (25)–(27), i.e., a local bidifferential operator supported on the diagonal with total derivative order bounded by the degree of divergence of the unextended kernel. The coefficients are locally and covariantly constructed (curvature polynomials on curved spacetime).*

Sketch. The statement is a direct consequence of the microlocal spectrum condition and scaling-degree bounds for the distributions entering T_n , combined with the structure theorem for extensions (Section 2.4) and the naturality requirement under embeddings [6,8,9]. $\square$

4.3. Dirac Fields: Currents and Spinor Bilinears

Let $\psi, \bar{\psi}$ be free Dirac fields on (M, g) (possibly coupled to smooth background gauge and Yukawa fields). The algebra of microcausal functionals and the classification of Wick powers/time-ordered products are developed in [32,33]. Denote by $S_F(x, y)$ the Dirac Feynman parametrix. Consider the vector current $J^\mu(x) := \bar{\psi}(x)\gamma^\mu\psi(x)$ and the spinor field $\psi(y)$; by fermionic Wick calculus,

$$T(J^\mu(x), \psi(y)) =: \bar{\psi}(x)\gamma^\mu\psi(x)\psi(y) : - \gamma^\mu S_F(x, y)\psi(y). \quad (28)$$

The second term exhibits a short-distance singularity of the universal Hadamard form; differentiation with respect to x followed by the $x \rightarrow y$ limit produces local, covariant contact terms. Changing the extension of (28) at the diagonal yields the general local form

$$\omega(J^\mu(f), \psi(g)) = \sum_{|\alpha| \leq 1} c_\alpha^\mu \int_M \partial^\alpha (f g)(x) \psi(x) \mathrm{dvol}_g(x), \quad c_\alpha^\mu \text{ locally covariant constants}, \quad (29)$$

and similarly for $T(\bar{\psi}(y), J^\mu(x))$. For current–current products,

$$T(J^\mu(x), J^\nu(y)) =: J^\mu(x)J^\nu(y) : + \mathrm{tr}(\gamma^\mu S_F(x, y)\gamma^\nu S_F(y, x)), \quad (30)$$

the extension ambiguity lies in the last trace term; a change in extension produces

$$\omega(J^\mu(f), J^\nu(g)) = \sum_{|\alpha| \leq 2} d_\alpha^{\mu\nu} \int_M \partial^\alpha (f g)(x) \mathbf{1} \mathrm{dvol}_g(x), \quad (31)$$

with $d_\alpha^{\mu\nu}$ being local and covariant (built from γ -matrix traces and background fields). The structures (29)–(31) are again local bidifferential operators supported on the diagonal and of finite order fixed by scaling degree [33]. They will feed directly into the vertex analysis and Ward identities (Section 5).

4.4. Curved Backgrounds and Curvature Dependence

On a general globally hyperbolic spacetime, Hadamard parametrices have the universal form

$$H_F(x, y) = \frac{U(x, y)}{\sigma_+(x, y)} + V(x, y) \log \sigma_+(x, y) + W(x, y),$$

with smooth U, V, W obtained locally and covariantly from the metric (and bundle data), and σ_+ the signed geodesic squared distance with $i0$ -prescription. Products such as $H_F(x, y)^2$ and $\text{tr}(\gamma^\mu S_F(x, y) \gamma^\nu S_F(y, x))$ admit scaling expansions whose extensions differ by *curvature polynomials* multiplying delta and its derivatives [34,35]. Consequently, the coefficients $C_\alpha^{(r)}$ in (20) and the constants (25), (29), (31) become curvature-dependent (e.g., involve $R, R_{\mu\nu}$ and their derivatives) with dimensions fixed by the scaling degree. This is the precise sense in which ω is *locally and covariantly* constructed.

4.5. Compatibility with the *-Structure and Microcausality

If Z is chosen to preserve causal unitarity (normalization N6) and field independence (N7), then Z_1 intertwines the *-operation and one has

$$\omega(A, B)^\dagger = \omega(B^\dagger, A^\dagger), \quad (32)$$

as required for $(A * B)^\dagger = B^\dagger * A^\dagger$ to hold at $O(\varepsilon)$ [9,16]. Microcausality is preserved to the first order because the support of ω is confined to diagonals (Proposition 3); hence, if $\text{supp } A \perp \text{supp } B$, then $A * B = B * A + O(\varepsilon^2)$.

4.6. Associativity Check on Generators

Although associativity to $O(\varepsilon)$ is guaranteed by $\delta\omega = 0$, it is instructive to verify it directly on generators. Let $F_i =: \phi^{n_i} : (f_i)$. Using (25)–(27) and the combinatorics of Wick contractions, it follows that

$$(F_1 * F_2) * F_3 - F_1 * (F_2 * F_3) = \varepsilon [\delta\omega](F_1, F_2, F_3) + O(\varepsilon^2) = O(\varepsilon^2),$$

because the only nonzero contributions at first order would arise from *two* contact singularities simultaneously—which are absent in the first-order perturbation of T_2 by construction (the T_3 ambiguity enters only at higher orders in the Stückelberg–Petermann expansion). The same holds in the Dirac sector using (29)–(31).

4.7. On the Nontriviality of $[\omega]$

Within the class of *local, covariant* 1-cochains $\eta : \mathcal{A} \rightarrow \mathcal{A}$ of bounded differential order (consistent with the normalization N0–N7), not every local contact shift (25)–(31) can be written as $\delta\eta$. For instance, the identity contribution in (25) proportional to $\int \partial^\alpha (fg)$ cannot be generated by a $\delta\eta$ that preserves the engineering dimensions of Wick polynomials and respects field independence. Indeed, $\delta\eta(\phi^2 : (f), \phi^2 : (g))$ is a linear combination of $\eta(\phi^2 : (fg))$ and $\phi^2 : (f)\eta(\phi^2 : (g)) + \eta(\phi^2 : (f))\phi^2 : (g)$, which, under the stated constraints, cannot reproduce a *pure multiple of the identity* with the derivative structure required by (25). A similar obstruction appears in the current sector (31), where the symmetric tensor $d_{\alpha}^{\mu\nu}$ defines a central term in the current algebra that cannot arise from a derivation-like η consistent with covariance and scaling. These observations, together with the classification results for Wick powers and time-ordered products [8,9,33], provide explicit examples where $[\omega] \neq 0$ in the subcategory of cochains compatible with the AQFT axioms. (We stress that this is a statement about nontriviality *within* the natural class of local, covariant cochains of bounded order; without such restrictions, continuous Hochschild

cohomology of certain C^* - or von Neumann completions may be trivial. Our setting is the perturbative, microlocal algebra of local fields.)

4.8. A No-Go Lemma for Trivialization by Local 1-Cochains

We now formalize the claim that the scalar and current central terms identified in Equations (25) and (31) cannot arise as Hochschild coboundaries $\delta\eta$ of any admissible local 1-cochain η compatible with our axioms. The proof isolates precise hypotheses on the allowed η and then uses a filtration by Wick degree together with the unit and field independence normalizations to rule out pure central outputs.

Set-up and notation.

Let $\mathcal{A}$ be the topological $*$ -algebra generated by Wick powers (scalar and Dirac sectors) on a globally hyperbolic spacetime, completed as in Section 2. Denote by $\deg W$ the Wick degree (number of fields in the normal-ordered monomial); for instance, $\deg:\phi^k:(f) = k$, $\deg J^\mu(f) = 2$, etc. Let $\pi_0 : \mathcal{A} \rightarrow \mathbb{C} \cdot \mathbf{1}$ be the projection onto the central (multiple-of-the-identity) component defined by the vacuum two-point structure used for normal ordering (equivalently, the map extracting the coefficient of $\mathbf{1}$ in the Wick expansion). For $A, B \in \mathcal{A}$, write

$$(\delta\eta)(A, B) = \eta(AB) - \eta(A)B - A\eta(B).$$

Admissible local 1-cochains.

We call a linear map $\eta : \mathcal{A} \rightarrow \mathcal{A}$ *admissible* if it satisfies

- (L1) **Locality and covariance:** η is given on local generators by a *finite* sum of differential operators on the test function jets with locally covariant coefficient tensors (as in (20)). In particular, $\text{supp } \eta(F)$ is contained in $\text{supp } F$ and η is natural under isometric embeddings.
- (L2) **Unit and field independence:** $\eta(\mathbf{1}) = 0$ and η commutes with functional differentiation on microcausal functionals in the sense induced by the normalization N7 of time-ordered products (no explicit dependence on external test functions beyond the functional's integrands).
- (L3) **Filtration (no degree drop to 0):** For any W with $\deg W \geq 1$, the central component of $\eta(W)$ vanishes, i.e., $\pi_0\eta(W) = 0$. Equivalently, $\deg \eta(W) \geq 1$ whenever $\deg W \geq 1$.
- (L4) **Scaling bound:** η respects the scaling-degree bounds implied by the EG extension at the order considered (so the total derivative order is bounded by the degree of divergence, as in Proposition 3).

Conditions (L1)–(L2) are the 1-cochain analogues of our renormalization axioms (N0–N7), in particular, the unit and field independence normalizations ensure that η annihilates constants and introduces no spurious nonlocal dependence. (L3) encodes that admissible finite renormalizations cannot turn *nontrivial* Wick polynomials into pure c -numbers; it is the standard filtration property for local redefinitions consistent with the Hollands–Wald classification of Wick polynomials. (L4) matches the microlocal scaling bounds.

Lemma 1 (No central coboundary from admissible η). *Let η be an admissible local 1-cochain. Then, for any $A, B \in \mathcal{A}$ with $\deg A \geq 1$ and $\deg B \geq 1$,*

$$\pi_0((\delta\eta)(A, B)) = 0. \quad (33)$$

In particular, for $A = :\phi^2:(f)$ and $B = :\phi^2:(g)$, one has

$$\pi_0((\delta\eta)(A, B)) = 0,$$

so $(\delta\eta)(A, B)$ cannot reproduce the central contact structure of Equation (25). Likewise, for $A = J^\mu(f)$ and $B = J^\nu(g)$, one cannot obtain the central tensor structure of Equation (31) from any admissible η .

Proof. We take central parts with π_0 and use linearity:

$$\pi_0(\delta\eta(A, B)) = \pi_0\eta(AB) - \pi_0(\eta(A)B) - \pi_0(A\eta(B)).$$

By (L3), $\pi_0(\eta(A)) = 0$ and $\pi_0(\eta(B)) = 0$ for $\deg A, \deg B \geq 1$. Since multiplication by B (resp. A) cannot create a central component out of a noncentral element in the Wick filtration (equivalently, if X has vanishing central part, then so do XB and AX when $\deg B, \deg A \geq 1$, because any central component would require full Wick contraction with a c -number kernel independent of the fields, which is excluded by (L2)–(L3)), we get $\pi_0(\eta(A)B) = \pi_0(A\eta(B)) = 0$. Thus,

$$\pi_0(\delta\eta(A, B)) = \pi_0\eta(AB).$$

Decompose AB into its Wick expansion. Write $AB = (AB)_{\text{nc}} + (AB)_{\text{cen}} \mathbf{1}$, where $(AB)_{\text{nc}}$ has strictly positive Wick degree and $(AB)_{\text{cen}}$ is a local distribution built from the parametrix contractions (for the pairs considered, this is exactly the H_F^2 -type object whose EG extension produces the central contact terms). Applying η and using linearity,

$$\pi_0\eta(AB) = \pi_0(\eta((AB)_{\text{nc}})) + \pi_0(\eta((AB)_{\text{cen}} \mathbf{1})).$$

The first term vanishes by (L3), because $\deg(AB)_{\text{nc}} \geq 1$ implies $\pi_0\eta((AB)_{\text{nc}}) = 0$. For the second term, (L2) (unit and field independence) gives $\eta(\mathbf{1}) = 0$ and forbids a nontrivial action of η on pure c -number multiples of the unit beyond differential relabelings already absorbed in $(AB)_{\text{cen}}$; combined with (L1)–(L4), this yields $\pi_0\eta((AB)_{\text{cen}} \mathbf{1}) = 0$. Hence $\pi_0\eta(AB) = 0$, and (33) follows. $\square$

Remark 2 (Consequences for the cocycle class). Lemma 1 shows that the central bidistributions appearing in (25) and (31) cannot be written as $\delta\eta$ with η admissible. Therefore, within the local, covariant subcomplex singled out by (L1)–(L4), the cocycle ω obtained from EG extension differences has a nontrivial central projection in $HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$. In particular, the algebraic curvature class $[\omega]$ is nonzero in this subcomplex whenever those central pieces are present.

Remark 3 (On the hypotheses). Hypothesis (L3) is the filtration statement one expects from the Hollands–Wald classification of Wick polynomials: admissible finite renormalizations map Wick powers to linear combinations of Wick powers of nonnegative degree with local differential operators on the smearing and do not collapse a nontrivial Wick monomial into a pure c -number. Hypothesis (L2) imports the unit and field independence normalizations (N2, N7) at the level of η , ensuring $\eta(\mathbf{1}) = 0$ and excluding spurious test function dependence. Both are satisfied by the Stückelberg–Petermann maps used elsewhere in this section.

4.9. Summary

We have exhibited ω explicitly as a local, covariant bidifferential operator supported on diagonals, derived from the finite extension freedom in T_2 (and, mutatis mutandis, higher T_n). For scalar and Dirac fields, we displayed the structure of ω on a set of generators (Wick monomials and currents). The resulting cocycle preserves the $*$ -structure and microcausality to first order and is, under natural constraints, cohomologically nontrivial. These constructions form the concrete backbone for the Ward identity analysis and for the anomaly/cohomology bridge developed in the next section.

5. Ward–Takahashi Identities and Deformed Vertices

In this section, we establish that the deformation of the observable product

$$A * B = AB + \varepsilon \omega(A, B) + O(\varepsilon^2), \quad \delta\omega = 0,$$

constructed in Sections 3 and 4, is compatible with gauge-current conservation and Ward–Takahashi identities (WTIs) at first order in ε under natural locality and biderivation hypotheses on ω . We then compute the induced structure of the *vertex function* Γ^μ for a Dirac field minimally coupled to a $U(1)$ background, showing how the transverse (gauge-invariant) form factors are controlled by the tensorial decomposition of ω , while the longitudinal part remains fixed by the WTI to $O(\varepsilon)$. Finally, we comment on the non-Abelian/BRST generalization.

5.1. Setup: Background-Coupled Current and Relative S -Matrix

Let $J^\mu(x) =: \bar{\psi}(x)\gamma^\mu\psi(x)$: denote the conserved Noether current of a Dirac field for a $U(1)$ gauge symmetry in the free theory. We regard an external gauge potential A_μ as a classical background and introduce the coupling via the local functional

$$V_A := \int_M A_\mu(x) J^\mu(x) \, \mathrm{dvol}_g(x). \quad (34)$$

In the pAQFT framework reviewed in Section 2, interacting (retarded) fields are defined with the relative S -matrix

$$S(V) := T \exp\left(\frac{i}{\hbar} V\right), \quad S_V(F) := S(V)^{-1} \star S(V + F), \quad R_V(F) := \left. \frac{\mathrm{d}}{\mathrm{d}\lambda} \right|_{\lambda=0} S_V(\lambda F),$$

where T is a fixed time-ordered product obeying the axioms. Gauge variations $\delta_\alpha A_\mu = \partial_\mu \alpha$ induce functional identities for $S(V_A)$ that are equivalent to the WTIs for time-ordered products with current insertions [7,16].

We now *change the product* on the algebra of microcausal functionals by the deformation $*$ induced by a scheme change $Z \in \mathrm{SP}$, as in (21). We write T_* for the time-ordered maps constructed from the same renormalization prescription but with a product $*$ used to concatenate factors (equivalently, we transport the old T through Z ; all constructions remain local and covariant at $O(\varepsilon)$). We emphasize that T_* and the associated retarded products differ from T by local contact terms controlled by ω and preserve causal factorization at first order.

5.2. Master Ward Identity at First Order

Let $X \equiv F_1 \otimes \cdots \otimes F_n$ be a tensor of local functionals with pairwise disjoint supports. In the undeformed theory, one has the familiar current Ward identity (distributionally)

$$\partial_\mu^x T(J^\mu(x), X) = i \sum_{k=1}^n \delta(x - x_k) T(F_1, \dots, \delta^g F_k, \dots, F_n), \quad (35)$$

where δ^g is the infinitesimal gauge variation acting on F_k and $x_k \in \mathrm{supp} F_k$; the right-hand side collects the standard contact terms at coincident points [16,36,37]. We say that a bilinear map ω is a *biderivation modulo local contact terms* if it satisfies (18).

Proposition 4 (Master Ward Identity to $O(\varepsilon)$). Assume ω is local, supported on partial diagonals, and a biderivation modulo local contact terms. Then

$$\partial_\mu^x T_*(J^\mu(x), X) = i \sum_{k=1}^n \delta(x - x_k) T_*(F_1, \dots, \delta^g F_k, \dots, F_n) + \varepsilon \partial_\mu^x C^\mu(x; X) + O(\varepsilon^2), \quad (36)$$

where $C^\mu(x; X)$ is a local, covariant functional (polynomial in jets of the F_i) supported on the diagonals $x = x_k$. Equivalently, the violation of the naive Leibniz rule introduced by ω is absorbed into contact terms that preserve the distributional WTI structure.

Sketch. Write $T_* = T + \varepsilon \Delta T + O(\varepsilon^2)$ and expand the left-hand side. By Section 4, the first-order change ΔT is a finite sum of distributions supported on diagonals, with coefficients built locally and covariantly from the background. Differentiation with respect to x preserves this support. The biderivation property (18) ensures that away from diagonals (i.e., for mutually disjoint points), the Leibniz rule holds and the current remains conserved at order ε ; therefore, any deviation must be a sum of contact (delta and derivatives) terms, which we package into $\partial_\mu C^\mu$. Local covariance follows from that of Z and of the EG extension. $\square$

5.3. Normalization of Contact Terms in the Ward–Takahashi Identity

Proposition 4 showed that the deformed time-ordered products obey the Master Ward Identity up to a local contact insertion $\partial_\mu C^\mu(x; X)$ supported on diagonals. In the AQFT framework, one interprets these contact terms as *renormalization ambiguities of the current operator itself*, to be fixed by suitable normalization conditions. We now spell out precise hypotheses under which the offending term can be removed once and for all by current improvement and field renormalization.

Lemma 2 (Normalization of contact terms in the WTI). Let ω be a local Hochschild 2 cocycle satisfying the biderivation–mod–contacts property (18). Then there exist renormalizations

(i) A redefinition of the current by a local improvement term

$$J^\mu \mapsto J_{\text{imp}}^\mu := J^\mu + \partial_\nu B^{[\mu\nu]}, \quad B^{[\mu\nu]} \text{ local, antisymmetric, covariant}, \quad (37)$$

(ii) A finite wavefunction renormalization of the Dirac field,

$$\psi \mapsto Z_\psi^{1/2} \psi, \quad Z_\psi = 1 + \varepsilon z + O(\varepsilon^2), \quad z \in \mathbb{R}, \quad (38)$$

such that the contact term $\partial_\mu C^\mu$ in (36) vanishes identically at $O(\varepsilon)$, i.e., the renormalized current J_{imp}^μ and field ψ satisfy the clean Ward–Takahashi identity

$$\partial_\mu^x T_*(J_{\text{imp}}^\mu(x), X) = i \sum_{k=1}^n \delta(x - x_k) T_*(F_1, \dots, \delta^g F_k, \dots, F_n) + O(\varepsilon^2), \quad (39)$$

for all local tensors $X = F_1 \otimes \dots \otimes F_n$.

Sketch of proof. The local contact term $\partial_\mu C^\mu(x; X)$ obtained in Proposition 4 is a sum of distributions supported on the diagonals with coefficients that are local polynomials in the background fields and jets of the F_i . By covariance, they can only arise from (i) improvement terms of the current ($\partial_\nu B^{[\mu\nu]}$) and (ii) local multiplicative renormalization of the matter field entering $J^\mu = \bar{\psi} \gamma^\mu \psi$.

Indeed, (i) accounts for derivative-type contact terms proportional to total derivatives of delta functions. Choosing $B^{[\mu\nu]}$ appropriately absorbs these into the definition of the

current. (ii) accounts for contact terms proportional to $\delta(x - x_k)\psi(x)$ or $\delta(x - x_k)\bar{\psi}(x)$, which are removed by a finite rescaling of ψ . Since ω is local and a biderivation modulo contacts, no further structures are possible at $O(\varepsilon)$. Thus, all contact contributions can be absorbed into the pair $(J_{\text{imp}}^\mu, Z_\psi)$, yielding the clean WTI (39). $\square$

Remark 4. This is the rigorous AQFT analogue of the usual textbook statement that conservation of the renormalized current can be restored by a suitable choice of finite counterterms and field renormalization. Here it appears as a structural consequence of locality and the biderivation property of ω .

Corollary 1 (Charge action as a derivation to first order). Let $Q(f) := \int J^0(x)f(x) d^3x$ with f compactly supported on a Cauchy surface. Then for observables A, B localized in the support of f , one has

$$\frac{i}{\hbar}[Q(f), A * B] = \left(\frac{i}{\hbar}[Q(f), A]\right) * B + A * \left(\frac{i}{\hbar}[Q(f), B]\right) + O(\varepsilon^2), \quad (40)$$

i.e., the charge acts by derivations on $(\mathcal{A}, *)$ up to $O(\varepsilon^2)$ (and local contact terms).

This corollary encodes the algebraic content of the WTI at first order: the symmetry is compatible with the deformed product to the extent that ω is a biderivation modulo local counterterms.

5.4. Momentum Space WTI and the Vertex Decomposition

We now pass to momentum space in Minkowski spacetime to extract the vertex function. Define the Fourier transform conventions

$$\tilde{F}(p) = \int d^4x e^{ip \cdot x} F(x), \quad F(x) = \int \frac{d^4p}{(2\pi)^4} e^{-ip \cdot x} \tilde{F}(p).$$

Let $S_*(p)$ be the exact (two-point) propagator in the deformed theory, i.e., the Fourier transform of $\langle T_* \psi(x) \bar{\psi}(y) \rangle$, and let $\Gamma_*^\mu(p', p)$ be defined by the amputated three-point function with one current insertion,

$$(2\pi)^4 \delta(p' + q - p) S_*^{-1}(p') G_*^\mu(p', p) S_*^{-1}(p) = \Gamma_*^\mu(p', p), \quad G_*^\mu(x, y; z) := \langle T_* \psi(x) \bar{\psi}(y) J^\mu(z) \rangle.$$

Transforming (36) and using standard manipulations yields the momentum-space WTI

$$q_\mu \Gamma_*^\mu(p', p) = S_*^{-1}(p') - S_*^{-1}(p) + \varepsilon \Xi(p', p; q) + O(\varepsilon^2), \quad (41)$$

where Ξ is a polynomial in external momenta originating from the Fourier transform of $\partial \cdot C$. As it is a *local* polynomial, it can be absorbed into a redefinition of local counterterms (field renormalizations and improvement terms in the current). Upon imposing a standard renormalization condition for the current, we set $\Xi = 0$ henceforth and obtain the familiar WTI to $O(\varepsilon)$:

$$q_\mu \Gamma_*^\mu(p', p) = S_*^{-1}(p') - S_*^{-1}(p) + O(\varepsilon^2). \quad (42)$$

The most general Lorentz-covariant decomposition of the vertex for a Dirac field reads

$$\Gamma_*^\mu(p', p) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m} F_2(q^2) + q^\mu \not{q} F_3(q^2) + \gamma^\mu \gamma_5 F_A(q^2) + \frac{\sigma^{\mu\nu} q_\nu}{2m} \gamma_5 F_{\text{EDM}}(q^2), \quad (43)$$

with $q = p' - p$, and $F_i = F_i^{(0)} + \varepsilon \Delta F_i + O(\varepsilon^2)$. The WTI (42) constrains the *longitudinal* piece:

$$F_1(q^2) - 1 \text{ and } F_3(q^2) \text{ are fixed by } S_*^{-1}(p) \text{ via } q_\mu \Gamma_*^\mu, \quad (44)$$

while the transverse form factors F_2, F_A, F_{EDM} are unconstrained by (42) and can receive contributions linear in ω .

5.5. How ω Feeds the Transverse Form Factors

The explicit local forms (29)–(31) imply that the first-order change of three-point functions with one current insertion is a sum of *contact* contributions and terms with the tensor structures of Pauli and electric-dipole operators. Schematically, from (29), one finds (suppressing polynomial contact pieces)

$$\Delta\Gamma^\mu(p', p) = \varepsilon \left[a(q^2) \gamma^\mu + b(q^2) \frac{i\sigma^{\mu\nu} q_\nu}{2m} + c(q^2) q^\mu \not{q} + d(q^2) \gamma^\mu \gamma_5 + \tilde{d}(q^2) \frac{\sigma^{\mu\nu} q_\nu}{2m} \gamma_5 \right], \quad (45)$$

with scalar form factors $a, b, c, d, \tilde{d}$ determined by the ω -coefficients c^μ_α and d^μ_α after amputation and renormalization. The WTI (42) enforces the relation between a, c and the self-energy; thus, the *independent* deformations at the first order reside in

$$\Delta F_2(q^2) = b(q^2), \quad \Delta F_A(q^2) = d(q^2), \quad \Delta F_{\text{EDM}}(q^2) = \tilde{d}(q^2).$$

In particular, the *anomalous magnetic moment* shift is

$$\Delta a_\ell \equiv \Delta F_2^{(\ell)}(0) = \varepsilon b_\ell(0), \quad (46)$$

and (if CP is violated by the ω -tensor structure) the *electric dipole moment* is

$$d_\ell = \frac{e}{2m_\ell} \varepsilon \tilde{d}_\ell(0). \quad (47)$$

Equations (46) and (47) provide the desired linear map from the algebraic cocycle data to measurable transverse vertex deformations.

5.6. CP, P, and T Properties of ω

Decompose the cocycle into CP-even and CP-odd parts, $\omega = \omega^{(+)} + \omega^{(-)}$, according to its action on spinor bilinears. For instance, an ω proportional to $\bar{\psi} \sigma^{\mu\nu} \psi F_{\mu\nu}$ is CP-even, while a component proportional to $\bar{\psi} \sigma^{\mu\nu} \gamma_5 \psi F_{\mu\nu}$ is CP-odd. Then,

$$\omega^{(+)} \Rightarrow \Delta F_2 \neq 0, \quad \Delta F_{\text{EDM}} = 0; \quad \omega^{(-)} \Rightarrow \Delta F_{\text{EDM}} \neq 0,$$

to the first order. This dichotomy follows from the transformation properties of the Dirac bilinears and the fact that the WTI constrains only the longitudinal part, leaving the transverse CP-odd/even sectors available for deformation. The local, covariant structure of ω (Section 4) ensures that these statements are independent of the regularization scheme.

5.7. Example: Contact Terms and On-Shell Normalization

To make the handling of the Ξ -term in (41) explicit, consider the typical contact contribution arising from (29) with $|\alpha| = 1$:

$$\omega(J^\mu(f), \psi(g)) \supset c^\mu_\nu \int \partial^\nu(fg)(x) \psi(x) \text{dvol}_g(x).$$

Fourier transforming and amputating yields a term in Γ^μ proportional to $c^\mu_\nu q^\nu$. Its divergence $q_\mu c^\mu_\nu q^\nu$ is a *polynomial* in external momenta and can be absorbed by imposing the on-shell renormalization condition $F_1(0) = 1$ and by choosing a conserved improved current (if desired) via the usual freedom to add total derivatives to J^μ . This implements $\Xi \rightarrow 0$ in (41) without affecting the transverse form factors.

5.8. Comparison with Standard Star-Product Models

In Moyal-type noncommutative QFT, the star product is defined from a constant Poisson bivector and produces momentum-dependent phases in vertices while preserving the $U(1)$ Ward identity because derivations with respect to the Moyal product are inner and satisfy Leibniz rules [26,29]. In our setting, the deformation arises from the Stückelberg–Petermann torsor and is *local* (supported on diagonals); nevertheless, the biderivation property (18) plays the same structural role and suffices to ensure the WTI to $O(\varepsilon)$ after appropriate normalization. The crucial difference is that our ω is not tied to a global Poisson structure but to renormalization ambiguities organized cohomologically; its physical imprint is therefore concentrated in transverse, local operator mixings (Pauli and EDM) rather than in nonlocal phase factors.

5.9. BRST/Slavnov–Taylor Generalization

For non-Abelian gauge theories, the WTIs are replaced by Slavnov–Taylor (ST) identities, most naturally encoded in the BRST framework. Let s be the BRST differential acting on the algebra of fields and antifields. In the pAQFT setting, the Master Ward Identity (MWI) expresses s -invariance of the generating functional up to local anomalies characterized by the BRST cohomology in ghost number one [7,38]. If ω is chosen to respect the biderivation property with respect to the BRST differential (modulo local counterterms), then the ST identity holds to $O(\varepsilon)$ with possible local breaking terms entirely controlled by the same cohomology class that will appear in the anomaly bridge of the next section. Thus, the present Abelian analysis extends verbatim to the BRST setting at the first order.

5.10. Summary

We have shown that (i) the current WTIs survive the first-order product deformation induced by ω , up to local contact terms that can be fixed by normalization, and (ii) the transverse part of the vertex—which encodes the Pauli and electric-dipole operators—receives linear, calculable corrections determined by the tensorial decomposition of ω . Equations (42) and (43) and the maps (46) and (47) provide the algebraic bridge between cohomological curvature and measurable low-energy form factors.

6. Anomalies from Algebraic Curvature: The BRST/BV Bridge

In this section, we connect the first-order deformation class $[\omega] \in HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$ of Sections 3 and 4 with the standard cohomological classification of anomalies. We work in the BRST/BV framework and the perturbative-algebraic setting of time-ordered products. Our main statements are (i) the deformed product $*$ induces, via the (anomalous) Master Ward Identity, a local anomaly insertion A_ω that solves the Wess–Zumino (WZ) consistency condition; (ii) the assignment $[\omega] \mapsto [a_\omega]$ defines a canonical homomorphism

$$\Phi : HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) \longrightarrow H^{1,d}(s|d),$$

where $H^{1,d}(s|d)$ is the local BRST cohomology (ghost number 1, form degree d); (iii) Φ intertwines trivial deformations with trivial anomalies, and is compatible with renormalization-group redefinitions; (iv) in examples, the image of Φ reproduces the known gauge (ABJ), mixed, and Weyl anomalies.

6.1. Local BRST Bicomplex and Wess–Zumino Consistency

Let $\mathcal{V}$ denote the graded-commutative algebra of local forms built from fields, antifields, ghosts and finitely many of their derivatives (jets). The BRST differential s increases the ghost number by +1 and implements gauge symmetry; the exterior derivative d in-

creases the form degree by +1. The pair (s, d) defines a bicomplex, and anomalies are classified by the local cohomology

$$H^{1,d}(s|d) = \frac{\{a^{1,d} \in \mathcal{V} : sa^{1,d} + db^{2,d-1} = 0\}}{\{a^{1,d} \sim a^{1,d} + sm^{0,d} + dn^{1,d-1}\}},$$

where $a^{1,d}$ is the *consistent anomaly density* and $b^{2,d-1}$ the associated *descent form* [39]. Equivalently, if W is the generating functional, the WZ condition reads

$$\delta_{\alpha_1} \mathcal{A}(\alpha_2) - \delta_{\alpha_2} \mathcal{A}(\alpha_1) = \mathcal{A}([\alpha_1, \alpha_2]),$$

encoding $sa^{1,d} + d(\dots) = 0$ and defining $[a^{1,d}] \in H^{1,d}(s|d)$ [4,40–42]. In even spacetime dimension $d = 2n$, solutions descending nontrivially are generated by the Chern–Weil polynomial I_{2n+2} through the Stora–Zumino descent; for conformal (Weyl) symmetry, the solutions split into type A (Euler density) and type B (Weyl invariants) [43].

6.2. Anomalous Master Ward Identity in pAQFT

Let T be a renormalized family of time-ordered products fulfilling the pAQFT axioms, and let T_* be the corresponding family for the deformed product $A * B = AB + \varepsilon \omega(A, B) + O(\varepsilon^2)$. The (anomalous) Master Ward Identity (MWI) expresses the BRST variation of time-ordered products in terms of local insertions [44,45]:

$$sT_*(X) = \frac{i}{\hbar} T_*(sX) + \frac{i}{\hbar} T_*(\mathcal{A}_*(X)), \quad (48)$$

where X is a tensor of local functionals and $\mathcal{A}_*$ is a *local, covariant* map supported on diagonals. At the first order in ε , one has

$$\mathcal{A}_*(X) = \varepsilon \mathcal{A}_\omega(X) + O(\varepsilon^2), \quad \mathcal{A}_\omega \text{ local, polynomial in jets.}$$

Nilpotency of s together with causal factorization implies the WZ condition for $\mathcal{A}_\omega$; thus, the coefficient density $a_\omega^{1,d}$ defined by pairing $\mathcal{A}_\omega$ with background sources is a representative of a class in $H^{1,d}(s|d)$.

6.3. Constructing the Anomaly from the Cocycle

Let $\omega \in C^2(\mathcal{A}, \mathcal{A})$ be a *local* Hochschild 2-cocycle ($\delta\omega = 0$) arising from a Stückelberg–Petermann change of scheme (Sections 4 and 5). For a collection $X \equiv F_1 \otimes \dots \otimes F_n$ of local functionals with mutually disjoint supports, one defines

$$\mathcal{A}_\omega(X) := \sum_{k=1}^n (-1)^{\epsilon_k} \left(\omega(F_k, \mathbf{1}) \bullet_k X - \omega(\mathbf{1}, F_k) \bullet_k X \right) + \sum_{i < j} \Xi_{ij}^\omega(F_i, F_j) \bullet_{ij} X, \quad (49)$$

where $\bullet_k$ and $\bullet_{ij}$ denote the obvious insertions at the k -th (or i, j -th) slot and Ξ_{ij}^ω collects the contact terms generated by the biderivation property of ω (Section 4). (Equation (49) is the abstract form of the first-order contact insertions computed in Section 4; for explicit scalar and Dirac examples see (25)–(31).) Transporting (49) through the T_* -map yields the insertion in (48).

Proposition 5 (WZ consistency from Hochschild closedness). *Assume ω is local, covariant, and a biderivation modulo local contact terms. Then the associated insertion $\mathcal{A}_\omega$ satisfies*

$$s\mathcal{A}_\omega(X) = -d\mathcal{B}_\omega(X),$$

for some local $\mathcal{B}_\omega$, i.e., the corresponding density $a_\omega^{1,d}$ obeys $sa_\omega^{1,d} + db_\omega^{2,d-1} = 0$. In particular, $[\omega]$ determines a well-defined class $\Phi([\omega]) = [a_\omega] \in H^{1,d}(s|d)$.

Sketch. By (48), $s^2T_*(X) = 0$ yields

$$0 = \frac{i}{\hbar}T_*(s\mathcal{A}_\omega(X)) + \frac{i}{\hbar}T_*(\mathcal{A}_\omega(sX)).$$

Using locality/biderivation of ω and causal factorization, the second term is a total derivative acting on local polynomials built from the jets of X ; these assemble into $d\mathcal{B}_\omega(X)$. The first term then enforces $sa_\omega^{1,d} + db_\omega^{2,d-1} = 0$. The assumption $\delta\omega = 0$ is precisely the $O(\varepsilon)$ associativity that underlies the graded Jacobi identity needed to close the descent. $\square$

Proposition 6 (Trivial deformations give trivial anomalies). *If $\omega = \delta\eta$ is Hochschild-exact with η local and covariant, then $\mathcal{A}_\omega(X) = s\mathcal{M}_\eta(X) + d\mathcal{N}_\eta(X)$ for some local $\mathcal{M}_\eta, \mathcal{N}_\eta$, hence $\Phi([\omega]) = 0$ in $H^{1,d}(s|d)$.*

Sketch. Writing $\omega = \delta\eta$ in (49) and using the Leibniz rule for s together with causal factorization shows that $\mathcal{A}_\omega$ is a BRST coboundary modulo d of a local functional depending on η . $\square$

6.4. Definition of the Local Hochschild Complex and Construction of Φ

We now make precise the domain and codomain of the map

$$\Phi : HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) \longrightarrow H^{1,d}(s|d),$$

constructed informally in the previous subsections. The key ingredients are (i) the *local Hochschild cochain complex* for the algebra of observables in pAQFT and (ii) the anomalous Master Ward Identity that produces a BRST anomaly density from a Hochschild 2-cocycle. Local Hochschild complex.

Let $\mathcal{A}$ be the algebra generated by microcausal functionals on a globally hyperbolic spacetime (M, g) , as in Section 2. We define the graded vector space of *local Hochschild n -cochains* by

$$C_{\text{loc}}^n(\mathcal{A}, \mathcal{A}) := \left\{ \phi : \mathcal{A}^{\otimes n} \rightarrow \mathcal{A} \mid \phi \text{ is } n\text{-linear, continuous, and local/covariant} \right\},$$

where

- *Continuity* is with respect to the locally convex topology on $\mathcal{A}$ induced by the wave-front set seminorms on functional derivatives [6,7].
- *Locality* means, for $A_i \in \mathcal{A}$ with mutually disjoint supports, $\phi(A_1, \dots, A_n) = 0$; equivalently, ϕ is supported on the total diagonal $\{x_1 = \dots = x_n\}$ in configuration space.
- *Covariance* means ϕ is natural under embeddings of spacetimes $(M, g) \hookrightarrow (M', g')$, in the sense of locally covariant QFT [14]. In particular, $\alpha_h \circ \phi = \phi \circ \alpha_h^{\otimes n}$ for isometries h .

The Hochschild differential is the usual

$$(\delta\phi)(A_1, \dots, A_{n+1}) = A_1\phi(A_2, \dots, A_{n+1}) + \sum_{i=1}^n (-1)^i \phi(A_1, \dots, A_i A_{i+1}, \dots, A_{n+1}) + (-1)^{n+1} \phi(A_1, \dots, A_n) A_{n+1}.$$

It is straightforward to check that δ maps C_{loc}^n to C_{loc}^{n+1} , so $(C_{\text{loc}}^\bullet, \delta)$ is a subcomplex of the full Hochschild complex. We denote its cohomology by $HH_{\text{loc}}^\bullet(\mathcal{A}, \mathcal{A})$.

Definition of Φ .

Let $[\omega] \in HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$, with ω a local 2-cocycle ($\delta\omega = 0$). Construct the deformed product $A * B = AB + \varepsilon\omega(A, B) + O(\varepsilon^2)$ and corresponding deformed time-ordered products T_* by transport of structure. The anomalous Master Ward Identity (MWI) then produces, to first order in ε ,

$$sT_*(X) = \frac{i}{\hbar}T_*(sX) + \frac{i}{\hbar}T_*(\mathcal{A}_\omega(X)),$$

with $\mathcal{A}_\omega$ a local insertion supported on diagonals (Section 6.3). Pairing $\mathcal{A}_\omega$ with background ghost sources yields a local anomaly density $a_\omega^{1,d}$ of ghost number 1 and form degree d , i.e., a representative in $H^{1,d}(s | d)$. We define

$$\Phi([\omega]) := [a_\omega^{1,d}] \in H^{1,d}(s | d).$$

Well-definedness on cohomology.

Proposition 7. *The map Φ is well defined on cohomology:*

- (i) *If $\omega' = \omega + \delta\eta$ with $\eta \in C_{\text{loc}}^1(\mathcal{A}, \mathcal{A})$, then $a_{\omega'}^{1,d} = a_\omega^{1,d} + sm^{0,d} + dn^{1,d-1}$ for some local m, n , hence $\Phi([\omega']) = \Phi([\omega])$.*
- (ii) *If T and T' are two renormalization schemes related by a Stückelberg–Petermann map Z , their cocycles ω, ω' lie in the same cohomology class and $\Phi([\omega]) = \Phi([\omega'])$.*

The following commuting diagrams (Figures 1 and 2) summarize Proposition 7: independence of representative $\omega \mapsto \omega + \delta\eta$ and independence of renormalization scheme $T \mapsto T'$ within the Stückelberg–Petermann orbit.

$$\begin{array}{ccc} \omega & \xrightarrow{\text{MWI}} & a_\omega^{1,d} \\ \downarrow + \delta\eta & & \downarrow + sm^{0,d} + dn^{1,d-1} \\ \omega' = \omega + \delta\eta & \xrightarrow{\text{MWI}} & a_{\omega'}^{1,d} \end{array} \quad (\text{A})$$

$$\begin{array}{ccc} T & \xrightarrow{\omega(T)} & a_{\omega(T)}^{1,d} \\ \downarrow Z \in \text{SP} & & \parallel \\ T' & \xrightarrow{\omega(T')} & a_{\omega(T')}^{1,d} \end{array} \quad (\text{B})$$

Figure 1. Commutativity expressing the well-definedness of $\Phi : HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) \rightarrow H^{1,d}(s | d)$ on cohomology classes and its independence of the renormalization scheme. (A) Coboundary invariance. Changing the representative $\omega \mapsto \omega + \delta\eta$ shifts the anomaly density by BRST- and d -exact terms. Hence, $[a_{\omega'}^{1,d}] = [a_\omega^{1,d}]$ and $\Phi([\omega']) = \Phi([\omega])$. (B) Scheme invariance. If T' is obtained from T by a Stückelberg–Petermann map Z , then $\omega(T') = \omega(T) + \delta\eta$ in the local complex; therefore $[a_{\omega(T')}^{1,d}] = [a_{\omega(T)}^{1,d}]$ and $\Phi([\omega(T')]) = \Phi([\omega(T)])$.

$$\begin{array}{ccc} C_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) & \xrightarrow{\text{MWI}} & \Omega_{\text{loc}}^{1,d} / (s\text{-exact} + d\text{-exact}) \\ \downarrow \text{quotient by } B_{\text{loc}}^2 & & \downarrow \\ HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) & \xrightarrow{\Phi} & H^{1,d}(s | d) \end{array}$$

Figure 2. The map Φ factors through the quotient by local Hochschild coboundaries on the domain and by BRST- and d -exact representatives on the codomain; the top arrow is induced by the anomalous MWI.

Sketch. (i) If $\omega' = \omega + \delta\eta$, then the corresponding insertion $\mathcal{A}_{\omega'}$ differs from $\mathcal{A}_\omega$ by s - and d -exact local terms (see Proposition 6), so they define the same class in $H^{1,d}(s | d)$. (ii) The Main Theorem of Renormalization (Section 2.6) shows T' is obtained from T by a local SP map. At first order, this changes ω by a coboundary in C_{loc}^2 and hence does not affect Φ . $\square$

Homomorphism property.

Proposition 8. *The map Φ is a homomorphism of Abelian groups: $\Phi([\omega_1] + [\omega_2]) = \Phi([\omega_1]) + \Phi([\omega_2])$.*

Proof. Linearity in ω of both the deformation $A * B = AB + \varepsilon\omega(A, B)$ and the induced insertion $\mathcal{A}_\omega$ implies $a_{\omega_1+\omega_2}^{1,d} = a_{\omega_1}^{1,d} + a_{\omega_2}^{1,d}$ up to s - and d -exact terms, so cohomology classes add linearly. $\square$

Functoriality under embeddings.

Proposition 9. *The assignment $(M, g) \mapsto \Phi([\omega]_{(M,g)})$ is a natural transformation with respect to embeddings in the category **Loc** of globally hyperbolic spacetimes. In particular, if $\chi : (M, g) \hookrightarrow (M', g')$ is an embedding, then*

$$\alpha_\chi^* \Phi([\omega]_{(M',g')}) = \Phi([\omega]_{(M,g)}),$$

where α_χ is the induced morphism of observable algebras.

Sketch. Locality and covariance of ω (by definition of $C_{\text{loc}}^\bullet$) imply naturality of the deformation under embeddings. Since the anomalous MWI is functorial in the choice of background, the anomaly density $a_\omega^{1,d}$ transforms covariantly as well. Thus, Φ commutes with α_χ . $\square$

Summary.

We have now defined Φ rigorously, proved that it is well defined on local Hochschild cohomology, linear, and functorial with respect to spacetime embeddings. This makes precise the central claim of this section: *an intrinsic algebraic curvature class $[\omega] \in HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$ canonically determines a BRST anomaly class in $H^{1,d}(s \mid d)$.*

6.5. Gauge, Mixed, and Weyl Anomalies

Abelian and non-Abelian gauge anomalies.

For a chiral Dirac field, the current sector of ω (Section 4.3) produces, via (49), a density $a_\omega^{1,d}$ proportional to the invariant polynomial $\text{tr}(C F^n)$ in $d = 2n$ (with ghost C and curvature F), i.e., the standard consistent anomaly class. Adding the Bardeen–Zumino polynomial corresponds to changing the representative by d -exact terms, leaving the cohomology class unchanged [40]. The Fujikawa Jacobian realizes the same class at the path-integral level [46,47]. In particular, $\Phi([\omega])$ vanishes precisely when the usual representation-theoretic conditions for anomaly cancellation hold (trace constraints on cubic Casimirs, etc.) [39].

Mixed gauge–gravitational and pure gravitational anomalies.

When background curvature is present, the local, covariant coefficients in ω (Sections 4.3 and 4.4) generate mixed terms proportional to $\text{tr}(C) \text{tr}(R^n)$ in $d = 2n$; in even dimensions of the form $4k+2$, one also finds pure gravitational anomalies for chiral matter, in agreement with the index-theoretic classification [5,39].

Weyl (trace) anomalies.

Taking the BRST differential to be the Weyl differential s_W , the same construction yields $a_\omega^{1,d} = \sigma(\alpha_A E_d + \sum_i \beta_i I_i)$, where σ is the Weyl ghost, E_d the Euler density (type A) and I_i local Weyl invariants (type B), with coefficients determined by the curvature-dependent part of ω [43]. Again, different renormalization schemes (i.e., different representatives of $[\omega]$) shift $a_\omega^{1,d}$ by s_W - and d -exact pieces without changing their class.

6.6. Descent and Index: Factoring Φ Through Characteristic Classes

Let $\mathfrak{Ch}$ denote the space of invariant polynomials (Chern character, Pontryagin classes). In even dimension $d = 2n$, the anomaly polynomial $I_{2n+2} \in \mathfrak{Ch}$ determines by descent a cocycle $a^{1,d}$ [41,42]. In our setting, the principal, curvature-dependent symbol of the local bidifferential operator ω defines an element $\sigma(\omega)$ of $\mathfrak{Ch}$ and the map Φ factors as

$$HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) \xrightarrow{\sigma} \mathfrak{Ch} \xrightarrow{\text{descent}} H^{1,d}(s|d),$$

up to s - and d -exact terms. This makes precise the statement that ω encodes the *local index density* data of the anomaly. The factorization is consistent with the BV formulation of the quantum master equation (QME) in pAQFT, where the anomaly is the obstruction to solving the renormalized QME and satisfies the same WZ condition [44,45,48].

6.7. Scheme Independence and One-Loop Exactness

Because Φ is defined on Hochschild *classes*, finite renormalizations (i.e., changes Z of scheme) cannot change $[a_\omega]$; they only modify representatives by s - and d -exact terms (Proposition 6). Consequently, standard nonrenormalization results for consistent anomalies (e.g., Adler–Bardeen-type statements) are compatible with our picture: higher-order scheme changes shift $a^{1,d}$ by trivial cocycles but cannot alter their cohomology class [39]. This is mirrored on the MWI side, where violations can be removed by finite counterterms whenever the corresponding class vanishes; otherwise, an *anomalous MWI* holds with $a^{1,d}$ fixed [45].

6.8. Examples: Recovering Known Anomaly Structures

ABJ anomaly in $d = 4$.

Take chiral fermions in a representation R of a compact gauge group. The current-sector cocycle ω of (31) produces, upon applying Φ , the density

$$a_\omega^{1,4} = \frac{1}{24\pi^2} \text{tr}_R(C F \wedge F) + s(\cdots) + d(\cdots),$$

in agreement with consistent vs. covariant choices distinguished by Bardeen–Zumino polynomials [40,46,47].

Weyl anomaly in $d = 4$.

For a conformally coupled scalar or a Dirac field, the curvature-dependent part of ω (Section 4.4) yields

$$a_\omega^{1,4} = \sigma(\alpha_A E_4 + \beta W^2) + s_W(\cdots) + d(\cdots),$$

with coefficients α_A, β determined by the field content; the split into type A/B follows the geometric classification of [43].

6.9. Summary

The first-order, local associative deformation encoded by $[\omega] \in HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$ determines, through the anomalous MWI, a BRST anomaly class $\Phi([\omega]) \in H^{1,d}(s|d)$. Hochschild closedness $\delta\omega = 0$ yields WZ consistency, while Hochschild exactness $\omega = \delta\eta$ produces a BRST-trivial anomaly. The curvature dependence of ω recovers the known geometric (index/characteristic-class) structure of gauge, mixed, and Weyl anomalies. Thus, the “algebraic curvature” $[\omega]$ provides an intrinsic AQFT origin for the anomaly cohomology and its observable consequences.

7. Phenomenological Applications: Calibration and Correlation Laws

In this section, we convert the first-order associative deformation

$$A * B = AB + \varepsilon \omega(A, B) + O(\varepsilon^2), \quad \delta\omega = 0,$$

into quantitative, low-energy predictions. The analysis proceeds in three steps: (i) an *effective-operator* matching of the tensor structures induced by ω onto a standard low-energy basis for lepton–photon couplings; (ii) an *RG evolution* from a matching scale to the experimental scale; (iii) *correlation laws* among observables obtained by fixing the single small parameter ε in one channel and propagating to others. Throughout, we retain only the leading (linear) dependence on ε and impose the Ward–Takahashi identities as derived in Section 5, so that only *transverse* (gauge-invariant) form factors receive independent deformations.

Here, ε is a formal bookkeeping parameter that tracks the first-order part of a scheme-induced deformation; its “smallness” reflects that we truncate the expansion at $O(\varepsilon)$ (with consistency controlled by $\delta\omega = 0$ and higher-order obstructions in HH^3 discussed in Section 8). The condition $\delta\omega = 0$ is not imposed by hand: it follows from associativity of the transported product and is the standard first-order Maurer–Cartan constraint in deformation theory.

7.1. Observables and Form-Factor Normalizations

For a charged Dirac lepton $\ell = e, \mu, \tau$, the $U(1)$ -covariant vertex function admits the decomposition

$$\Gamma^\mu(p', p) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m_\ell} F_2(q^2) + q^\mu \not{q} F_3(q^2) + \gamma^\mu \gamma_5 F_A(q^2) + \frac{\sigma^{\mu\nu} q_\nu}{2m_\ell} \gamma_5 F_{\text{EDM}}(q^2), \quad q = p' - p. \quad (50)$$

With on-shell renormalization, $F_1(0) = 1$, and the *anomalous magnetic moment* and *electric dipole moment* are

$$a_\ell \equiv F_2(0), \quad d_\ell \equiv \frac{e}{2m_\ell} F_{\text{EDM}}(0). \quad (51)$$

The Ward identity $q_\mu \Gamma^\mu = S^{-1}(p') - S^{-1}(p)$ fixes the longitudinal piece, leaving F_2, F_A, F_{EDM} unconstrained and therefore available to receive $O(\varepsilon)$ contributions from ω (Section 5). We focus on F_2 and F_{EDM} , which admit a compact effective-operator description.

7.2. From Algebraic Curvature to Effective Operators

The local, covariant cocycle ω constructed in Section 4 modifies time-ordered products with current insertions by contact terms that, after amputation, project onto the *electromagnetic dipole* structures. A convenient language is to match onto the SMEFT operator basis above the electroweak scale Λ :

$$\mathcal{L}_{\text{SMEFT}} \supset \frac{C_{e\gamma}^\ell}{\Lambda^2} (L_\ell \sigma^{\mu\nu} e_{R\ell}) H F_{\mu\nu} + \text{h.c.}, \quad (52)$$

with L_ℓ the lepton doublet, $e_{R\ell}$ the singlet, H the Higgs doublet, and $F_{\mu\nu}$ the electromagnetic field-strength. (We assume hypercharge/weak dipoles have been rotated to the mass basis so that (52) captures the photon coupling after EWSB.) Electroweak symmetry breaking (EWSB) $H \rightarrow (0, (v+h)/\sqrt{2})^T$ gives the LEFT operator

$$\mathcal{L}_{\text{LEFT}} \supset \frac{1}{2} \bar{\ell} \sigma^{\mu\nu} (c_\ell + i \tilde{c}_\ell \gamma_5) \ell F_{\mu\nu}, \quad c_\ell + i \tilde{c}_\ell = \frac{v}{\sqrt{2} \Lambda^2} C_{e\gamma}^\ell, \quad (53)$$

so that, comparing with (51),

$$\Delta a_\ell = \frac{2m_\ell}{e} \Re c_\ell = \frac{2m_\ell}{e} \frac{v}{\sqrt{2}\Lambda^2} \Re C_{e\gamma}^\ell, \quad d_\ell = -\tilde{c}_\ell = -\frac{v}{\sqrt{2}\Lambda^2} \Im C_{e\gamma}^\ell. \quad (54)$$

In our framework, the Stückelberg–Petermann deformation induces these dipoles *linearly* in ε . It is therefore natural to write

$$C_{e\gamma}^\ell(\Lambda) = \varepsilon \widehat{C}_{e\gamma}^\ell(\Lambda), \quad (55)$$

with $\widehat{C}_{e\gamma}^\ell$ as a dimensionless “reduced” Wilson coefficient fixed by the local tensor content of ω in the current sector (cf. (29)–(31)). Combining (54) and (55),

$$\boxed{\Delta a_\ell = \varepsilon \mathcal{K}_\ell \Re \widehat{C}_{e\gamma}^\ell, \quad d_\ell = -\varepsilon \mathcal{K}_0 \Im \widehat{C}_{e\gamma}^\ell}, \quad \mathcal{K}_\ell := \frac{2m_\ell}{e} \frac{v}{\sqrt{2}\Lambda^2}, \quad \mathcal{K}_0 := \frac{v}{\sqrt{2}\Lambda^2}. \quad (56)$$

Equation (56) is the basic “algebra $\rightarrow$ EFT $\rightarrow$ observable” map at the matching scale.

7.3. RG Evolution from Λ to the Experimental Scales

The LEFT dipole coefficients run under QED/QCD renormalization. To leading-log accuracy,

$$\mu \frac{d}{d\mu} \begin{pmatrix} c_\ell \\ \tilde{c}_\ell \end{pmatrix} = \frac{1}{16\pi^2} \begin{pmatrix} \gamma_{cc} & 0 \\ 0 & \gamma_{\tilde{c}\tilde{c}} \end{pmatrix} \begin{pmatrix} c_\ell \\ \tilde{c}_\ell \end{pmatrix} + \frac{1}{16\pi^2} \sum_j \begin{pmatrix} \gamma_{c\leftarrow j} \\ \gamma_{\tilde{c}\leftarrow j} \end{pmatrix} C_j, \quad (57)$$

where C_j denote other LEFT/SMEFT Wilson coefficients that mix into the dipoles at one loop (four-fermion, Higgs–lepton, etc.). The solution is

$$\begin{pmatrix} c_\ell(\mu) \\ \tilde{c}_\ell(\mu) \end{pmatrix} = U(\mu, \Lambda) \begin{pmatrix} c_\ell(\Lambda) \\ \tilde{c}_\ell(\Lambda) \end{pmatrix} + \sum_j \int_{\ln \mu}^{\ln \Lambda} \frac{dt}{16\pi^2} U(\mu, e^t) \begin{pmatrix} \gamma_{c\leftarrow j} \\ \gamma_{\tilde{c}\leftarrow j} \end{pmatrix} C_j(e^t), \quad U(\mu, \Lambda) = \exp \left[\frac{1}{16\pi^2} \Gamma \ln \frac{\mu}{\Lambda} \right], \quad (58)$$

with $\Gamma = \text{diag}(\gamma_{cc}, \gamma_{\tilde{c}\tilde{c}})$. In many scenarios, the mixing terms are subleading and one may take

$$c_\ell(\mu) \simeq c_\ell(\Lambda) \left[1 + \frac{\gamma_{cc}}{16\pi^2} \ln \frac{\mu}{\Lambda} \right], \quad \tilde{c}_\ell(\mu) \simeq \tilde{c}_\ell(\Lambda) \left[1 + \frac{\gamma_{\tilde{c}\tilde{c}}}{16\pi^2} \ln \frac{\mu}{\Lambda} \right]. \quad (59)$$

All results below remain valid to this order by replacing $c_\ell, \tilde{c}_\ell$ in (56) with their running values at $\mu = m_\ell$ for a_ℓ and at the experimental scale for d_ℓ . (For a systematic SMEFT-to-LEFT evolution including operator mixing, see [49–52].)

7.4. Calibration of ε and Propagation

Concreteness and testability.

The purpose of this section is not to propose a specific ultraviolet completion but to extract *falsifiable correlations* that follow from the cohomological organization of first-order deformations once a single calibration is chosen. In particular, Equations (61) and (63) give parameter-free relations (up to flavor texture assumptions) that can be confronted with improved measurements of $\Delta a_e, \Delta a_\mu$ and EDM bounds.

Fix a reference channel, e.g., the muon anomalous magnetic moment. We use the experimentally measured muon anomalous magnetic moment as the calibration input, as reported by the BNL E821 experiment and the Fermilab Muon g–2 collaboration [53,54]. For the Standard Model prediction and the definition of the discrepancy, we refer to the

current community theory evaluations [55]. Define the (experiment–SM) discrepancy Δa_μ (we keep it symbolic). At the scale $\mu = m_\mu$,

$$\varepsilon = \frac{\Delta a_\mu}{\mathcal{K}_\mu \Re \widehat{C}_{e\gamma}^\mu(\mu)}. \quad (60)$$

Inserting (60) into (56) yields *parameter-free correlations*:

$$\Delta a_e = \Delta a_\mu \times \frac{\mathcal{K}_e}{\mathcal{K}_\mu} \frac{\Re \widehat{C}_{e\gamma}^e(\mu)}{\Re \widehat{C}_{e\gamma}^\mu(\mu)} = \Delta a_\mu \times \frac{m_e}{m_\mu} \frac{\Re \widehat{C}_{e\gamma}^e(\mu)}{\Re \widehat{C}_{e\gamma}^\mu(\mu)}, \quad (61)$$

$$d_\ell = -\frac{e}{2m_\ell} \Delta a_\mu \times \frac{\Im \widehat{C}_{e\gamma}^\ell(\mu)}{\Re \widehat{C}_{e\gamma}^\mu(\mu)} \frac{\mathcal{K}_0}{\mathcal{K}_\mu} = -\Delta a_\mu \times \frac{e}{2m_\ell} \frac{\Im \widehat{C}_{e\gamma}^\ell(\mu)}{\Re \widehat{C}_{e\gamma}^\mu(\mu)} \frac{1}{(2m_\mu/e)}. \quad (62)$$

In particular, (56) implies the *EDM–g–2 phase relation*

$$\boxed{\frac{d_\ell}{\Delta a_\ell} = \frac{e}{2m_\ell} \tan \phi_\ell, \quad \phi_\ell := \arg \widehat{C}_{e\gamma}^\ell(\mu),} \quad (63)$$

independent of Λ and ε . Thus, any nonzero phase in the ω -induced dipole immediately correlates d_ℓ with Δa_ℓ .

7.5. Flavor Textures and Universal Correlation Laws

Equation (61) still depends on the *flavor texture* of $\widehat{C}_{e\gamma}^\ell$. Two benchmark hypotheses capture broad classes of UV completions and are natural from the algebraic standpoint (they reflect whether the chirality flip is provided dominantly by the Higgs insertion or not):

Texture U (flavor-universal). $\widehat{C}_{e\gamma}^\ell(\mu) \simeq \widehat{C}(\mu)$ independent of ℓ . Then $\Delta a_\ell \propto m_\ell$ and $d_\ell \propto \text{const}$:

$$\frac{\Delta a_e}{\Delta a_\mu} = \frac{m_e}{m_\mu}, \quad \frac{d_e}{d_\mu} \simeq 1, \quad \frac{d_\ell}{\Delta a_\ell} = \frac{e}{2m_\ell} \tan \phi \quad (\phi \text{ common}).$$

Texture Y (Yukawa-aligned/MFV). $\widehat{C}_{e\gamma}^\ell(\mu) \propto y_\ell(\mu)$ with $y_\ell = \sqrt{2} m_\ell/v$. Then $\Delta a_\ell \propto m_\ell y_\ell \propto m_\ell^2$ and $d_\ell \propto y_\ell \propto m_\ell$:

$$\frac{\Delta a_e}{\Delta a_\mu} = \left(\frac{m_e}{m_\mu}\right)^2, \quad \frac{d_e}{d_\mu} = \frac{m_e}{m_\mu}, \quad \frac{d_\ell}{\Delta a_\ell} = \frac{e}{2m_\ell} \tan \phi \quad (\text{as in (63)}).$$

These follow directly from (56) using $\widehat{C}_{e\gamma}^\ell = \{\widehat{C}, \kappa y_\ell\}$ and are stable under leading-log running because the QED anomalous dimensions are flavor universal.

7.6. Rigorous Statements and Proofs

We formalize the above as two propositions.

Proposition 10 (Phase–ratio relation). *Let the deformed vertex satisfy the WTI of Section 5, and assume that the transverse dipole structures arise solely from the local cocycle ω via (55) with a single complex coefficient $\widehat{C}_{e\gamma}^\ell$ for flavor ℓ . Then, (63) holds, i.e.,*

$$\frac{d_\ell}{\Delta a_\ell} = \frac{e}{2m_\ell} \tan(\arg \widehat{C}_{e\gamma}^\ell),$$

independent of Λ , ε , and RG improvement at the leading log.

Proof. Equation (56) gives $\Delta a_\ell = \varepsilon \mathcal{K}_\ell \Re \widehat{C}_{e\gamma}^\ell$ and $d_\ell = -\varepsilon \mathcal{K}_0 \Im \widehat{C}_{e\gamma}^\ell$. Division yields $d_\ell / \Delta a_\ell = -(\mathcal{K}_0 / \mathcal{K}_\ell) \tan \phi_\ell = (e / 2m_\ell) \tan \phi_\ell$. The multiplicative RG factors in (59) cancel in the ratio because they are identical for c_ℓ and $\tilde{c}_\ell$ at the leading log. $\square$

Proposition 11 (Flavor-scaling laws). *Under Texture U (resp. Y), the ratios in Table 1 hold to the leading log.*

Table 1. Parameter-free correlations implied by the algebraic-curvature deformation, for two benchmark flavor textures of $\widehat{C}_{e\gamma}^\ell$.

Texture	$\frac{\Delta a_e}{\Delta a_\mu}$	$\frac{d_e}{d_\mu}$	$\frac{d_\ell}{\Delta a_\ell}$
U (flavor-universal)	$\frac{m_e}{m_\mu}$	1	$\frac{e}{2m_\ell} \tan \phi$
Y (Yukawa-aligned/MFV)	$\left(\frac{m_e}{m_\mu}\right)^2$	$\frac{m_e}{m_\mu}$	$\frac{e}{2m_\ell} \tan \phi$

Proof. In Texture U, $\widehat{C}_{e\gamma}^\ell$ is a common constant, so from (56) $\Delta a_\ell \propto m_\ell$, $d_\ell \propto \text{const}$, and the listed ratios follow. In Texture Y, $\widehat{C}_{e\gamma}^\ell \propto y_\ell \propto m_\ell$, giving $\Delta a_\ell \propto m_\ell^2$ and $d_\ell \propto m_\ell$. Universality of the anomalous dimensions ensures that running does not spoil ratios at the leading log. $\square$

7.7. Uncertainty Propagation and Inference of ε

Let Δa_μ be calibrated with uncertainty σ_μ . Denote by $R_{e/\mu}$ the predicted ratio $\Delta a_e / \Delta a_\mu$ in a chosen texture, and by ρ_ℓ the (possibly complex) ratio of reduced Wilson coefficients $\widehat{C}_{e\gamma}^\ell / \widehat{C}_{e\gamma}^\mu$. Then

$$\Delta a_e = R_{e/\mu} |\rho_e| \cos(\delta_e) \Delta a_\mu, \quad \sigma(\Delta a_e) = \sqrt{(R_{e/\mu} |\rho_e| \cos \delta_e)^2 \sigma_\mu^2 + \Delta a_\mu^2 \sigma^2(R_{e/\mu} |\rho_e| \cos \delta_e)}, \quad (64)$$

where $\delta_e = \arg(\rho_e)$ and σ includes theoretical systematics from RG evolution and matching. Analogously, EDM predictions read

$$d_\ell = \frac{e}{2m_\ell} \tan \phi_\ell \Delta a_\ell, \quad \sigma(d_\ell) = \left| \frac{e}{2m_\ell} \right| \sqrt{\tan^2 \phi_\ell \sigma^2(\Delta a_\ell) + \Delta a_\ell^2 \sec^4 \phi_\ell \sigma^2(\phi_\ell)}. \quad (65)$$

Given measurements (or upper bounds) on d_e and Δa_μ , one can infer ε and ϕ_e up to texture assumptions by combining (60) and (63).

7.8. Beyond Dipoles: Additional Transverse Structures

While the dipole operator dominates low-energy, spin-flip observables, other transverse pieces can arise from ω :

- *Anapole* form factor $F_A(q^2)$ (axialvector coupling), which contributes to parity-violating Möller and atomic observables. The cocycle components acting on axial currents generate local operators of the form $\bar{\ell} \gamma^\mu \gamma_5 \ell \partial^\nu F_{\mu\nu}$ in LEFT; these are suppressed by q^2 but can be relevant at Z-pole energies.
- *Four-fermion* operators mixing into $C_{e\gamma}$ under RG, e.g., $(\bar{\ell} \sigma^{\mu\nu} e)(\bar{f} \gamma_\nu f)$; their leading effect is encoded in the second line of (57) and is modeled by the same ε through the ω -tensor structure in the matter sector.

These additions preserve the correlation logic but introduce further (controlled) parameters if included in the same order.

7.9. Consistency Checks: Dimensions, Limits, and Decoupling

The map (56) is dimensionally consistent: $[a_\ell] = 1$, $[d_\ell] = M^{-1}$, and $\mathcal{K}_\ell \propto m_\ell/\Lambda^2$ while $\mathcal{K}_0 \propto 1/\Lambda^2$. In the decoupling limit $\Lambda \rightarrow \infty$ or $\varepsilon \rightarrow 0$, both Δa_ℓ and d_ℓ vanish linearly in ε (quadratically if Texture Y is enforced through Yukawa scaling). If the phase vanishes, $\phi_\ell \rightarrow 0$, then $d_\ell/\Delta a_\ell \rightarrow 0$ as in (63). These properties are invariant under reparametrizations of the renormalization scheme because ε multiplies a class $[\omega]$ and observables depend only on $[\omega]$ through $\widehat{C}_{e\gamma}^\ell$.

7.10. Illustrative Worked Example (Symbolic)

Assume Texture Y with a common complex proportionality κ , $\widehat{C}_{e\gamma}^\ell = \kappa y_\ell$. Calibrating on μ ,

$$\varepsilon = \frac{\Delta a_\mu}{\mathcal{K}_\mu \Re(\kappa y_\mu)} = \frac{\Delta a_\mu}{(2m_\mu/e)(v/\sqrt{2}\Lambda^2)\Re(\kappa\sqrt{2}m_\mu/v)} = \frac{\Delta a_\mu}{(2/e)(m_\mu^2/\Lambda^2)\Re\kappa}.$$

Then

$$\Delta a_e = \varepsilon \mathcal{K}_e \Re(\kappa y_e) = \Delta a_\mu \left(\frac{m_e}{m_\mu}\right)^2 \frac{\Re\kappa}{\Re\kappa} = \Delta a_\mu \left(\frac{m_e}{m_\mu}\right)^2,$$

and

$$d_e = -\varepsilon \mathcal{K}_0 \Im(\kappa y_e) = -\Delta a_\mu \frac{e}{2m_e} \tan\phi \left(\frac{m_e}{m_\mu}\right)^2, \quad \phi := \arg\kappa,$$

in agreement with Table 1 and (63).

7.11. Remarks on Hadronic and SM Backgrounds

The map (56) concerns *new* contributions induced by ω . Standard Model (SM) backgrounds (QED, electroweak, hadronic) are additive and can be subtracted in defining the discrepancies Δa_ℓ . The hadronic vacuum polarization/light-by-light pieces are independent of ε and enter only through the input value of Δa_μ used for calibration; see [55,56] for comprehensive SM accounts. Our correlation laws are thus insensitive to the internal organization of the SM piece, provided a common convention is used across flavors.

7.12. Application to the Muon $g-2$: Calibration and Predictions

The electron anomalous magnetic moment has been measured with extremely high precision; we refer to the most precise determinations in the literature [57,58].

We now specialize the general map of Equation (52) to the muon channel and propagate to other leptons. Define the (experiment–SM) discrepancy symbolically by Δa_μ (we do not use any external numeric input here). The calibration step is, at the scale $\mu = m_\mu$,

$$\varepsilon = \frac{\Delta a_\mu}{\mathcal{K}_\mu \Re \widehat{C}_{e\gamma}^\mu(\mu)}, \quad \mathcal{K}_\mu = \frac{2m_\mu}{e} \frac{v}{\sqrt{2}\Lambda^2}, \quad (66)$$

cf. Equation (56). Inserting this into Equation (52) gives parameter-free correlations:

$$\Delta a_e = \Delta a_\mu \frac{\mathcal{K}_e}{\mathcal{K}_\mu} \frac{\Re \widehat{C}_{e\gamma}^e(\mu)}{\Re \widehat{C}_{e\gamma}^\mu(\mu)} = \Delta a_\mu \frac{m_e}{m_\mu} \frac{\Re \widehat{C}_{e\gamma}^e(\mu)}{\Re \widehat{C}_{e\gamma}^\mu(\mu)}, \quad (67)$$

$$\Delta a_\tau = \Delta a_\mu \frac{\mathcal{K}_\tau}{\mathcal{K}_\mu} \frac{\Re \widehat{C}_{e\gamma}^\tau(\mu)}{\Re \widehat{C}_{e\gamma}^\mu(\mu)} = \Delta a_\mu \frac{m_\tau}{m_\mu} \frac{\Re \widehat{C}_{e\gamma}^\tau(\mu)}{\Re \widehat{C}_{e\gamma}^\mu(\mu)}. \quad (68)$$

For the EDMs, we use the phase–ratio relation (59):

$$\frac{d_\ell}{\Delta a_\ell} = \frac{e}{2m_\ell} \tan\phi_\ell, \quad \phi_\ell = \arg \widehat{C}_{e\gamma}^\ell(\mu). \quad (69)$$

Two benchmark flavor textures, both natural in the present framework, then, imply the following:

- **Texture U (flavor-universal):** $\widehat{C}_{e\gamma}^\ell = \widehat{C}$.

$$\Delta a_e = \Delta a_\mu \frac{m_e}{m_\mu}, \quad \Delta a_\tau = \Delta a_\mu \frac{m_\tau}{m_\mu}, \quad d_\ell = \frac{e}{2m_\ell} \Delta a_\ell \tan \phi.$$

- **Texture Y (Yukawa-aligned/MFV):** $\widehat{C}_{e\gamma}^\ell = \kappa y_\ell$ with $y_\ell = \sqrt{2} m_\ell / v$.

$$\Delta a_e = \Delta a_\mu \left(\frac{m_e}{m_\mu}\right)^2, \quad \Delta a_\tau = \Delta a_\mu \left(\frac{m_\tau}{m_\mu}\right)^2, \quad d_\ell = \frac{e}{2m_\ell} \Delta a_\ell \tan \phi.$$

These relations are *derived entirely within the paper* from Equations (52)–(59); no external numbers are needed. Importantly, the signs of Δa_e and Δa_μ may differ here (depending on the relative phase of $\widehat{C}_{e\gamma}^e$ and $\widehat{C}_{e\gamma}^\mu$), which will become a key discriminator below.

7.13. Vector-Portal (“Dark-Photon”) Foil: Loop Structure, Scaling, and Discriminants

As a foil, consider a kinetically mixed $U(1)_D$ vector A' of mass $m_{A'}$ that couples as $e \chi \bar{\ell} \gamma^\mu \ell A'_\mu$ after the diagonalization of kinetic terms (we reserve ε for our algebraic-curvature deformation and write χ for the kinetic-mixing parameter). The one-loop contribution to the lepton anomaly is (vector coupling)

$$\Delta a_\ell^{(A')} = \frac{\alpha \chi^2}{2\pi} \int_0^1 dx \frac{2x(1-x)^2}{(1-x)^2 + x r_\ell}, \quad r_\ell := \frac{m_{A'}^2}{m_\ell^2}. \quad (70)$$

This integral is positive and yields the familiar limits

$$\Delta a_\ell^{(A')} \simeq \frac{\alpha \chi^2}{2\pi} \quad (m_{A'} \ll m_\ell), \quad \Delta a_\ell^{(A')} \simeq \frac{\alpha \chi^2}{3\pi} \frac{m_\ell^2}{m_{A'}^2} \quad (m_{A'} \gg m_\ell). \quad (71)$$

At the leading order with a pure vector coupling, A' does *not* induce an EDM ($d_\ell = 0$), since there is no CP-odd phase (CP-odd EDMs may arise only if additional CP-violating interactions are introduced beyond minimal kinetic mixing).

Qualitative comparison to algebraic curvature.

- **Origin and scaling.** Curvature: $\Delta a_\ell = \varepsilon \mathcal{K}_\ell \Re \widehat{C}_{e\gamma}^\ell$ is *linear* in the deformation parameter and *model-independent* at low energies, Section 7. Dark photon: $\Delta a_\ell^{(A')} \propto \chi^2$ and depends on $m_{A'}/m_\ell$ via the loop (70).
- **Sign structure.** Curvature can accommodate either sign for Δa_ℓ (through $\Re \widehat{C}_{e\gamma}^\ell$), and different flavors can in principle have opposite signs. The minimal vector A' gives a *positive* contribution to all ℓ .
- **EDM correlation.** Curvature predicts $d_\ell/\Delta a_\ell = (e/2m_\ell) \tan \phi_\ell$ (Equation (59)), so a nonzero phase *forces* a correlated EDM. The minimal dark photon has $d_\ell = 0$ at one loop. Hence, any robust nonzero d_ℓ aligned with Δa_ℓ favors curvature over a pure dark photon.
- **Flavor correlations.** Curvature gives the mass ratio laws in Section 7.12 (Texture U or Y). For a dark photon, the flavor pattern is governed by r_ℓ : $\Delta a_e^{(A')}/\Delta a_\mu^{(A')} = f(r_e)/f(r_\mu)$ with f the loop integral in (70).

Discriminant table (derived properties).

	Algebraic Curvature (This Work)	Vector Portal A' (Foil)
Scaling in coupling	$\propto \varepsilon$	$\propto \chi^2$
EDM at leading order	$d_\ell / \Delta a_\ell = (e/2m_\ell) \tan \phi_\ell$	$d_\ell = 0$ (CP-even)
Sign of Δa_ℓ	model-dependent (both signs allowed)	positive (vector)
Flavor pattern	mass-ratio laws (U or Y textures)	loop $f(m_{A'}/m_\ell)$
On-shell signatures	none required (no mediator)	visible/missing-energy A' signals

Thus, a simultaneous fit to Δa_μ , Δa_e and EDM bounds, together with the absence of on-shell A' signals, naturally selects the algebraic curvature interpretation, whereas observation of a narrow vector resonance or of the loop pattern $f(m_{A'}/m_\ell)$ would point to the vector portal.

Notation note. We use ε exclusively for the algebraic curvature deformation and χ for kinetic mixing to avoid confusion.

8. Limitations and Extensions

This section clarifies the precise scope of our results, formulates higher-order obstructions in the deformation problem, and outlines controlled extensions in three directions: (i) net-level and wedge-local deformations; (ii) operadic/factorization algebra lifts of the construction; (iii) gravitational and cosmological sectors, including a consistent decoupling scheme. Where possible, we give proofs or proof sketches that make the logical dependence on locality, covariance, and scaling degree explicit.

8.1. Scope and Standing Assumptions

All constructions up to now are *formal*, to first order in a bookkeeping parameter ε , within the perturbative algebraic framework (pAQFT) on globally hyperbolic spacetimes. We have the following:

- A local, covariant associative product deformation $A * B = AB + \varepsilon \omega(A, B) + O(\varepsilon^2)$ with $\delta\omega = 0$;
- Preservation of the $*$ -structure and microcausality at $O(\varepsilon)$ under the hypotheses of Section 4;
- Compatibility with Ward–Takahashi identities (WTIs) at $O(\varepsilon)$ modulo local contact terms fixed by normalization, Section 5;
- A functorial map $[\omega] \mapsto [a_\omega] \in H^{1,d}(s|d)$ to local anomaly classes, Section 6.

The following subsections characterize how these statements generalize to higher orders, to other notions of locality (wedge locality), to higher-homotopy (operadic) structures, and to gravitational sectors.

8.2. Higher Orders: Maurer–Cartan Tower and Obstructions in HH^3

Let m denote the original associative product and recall the Gerstenhaber bracket $[\cdot, \cdot]_G$. A formal deformation

$$m_\varepsilon \equiv m + \varepsilon \omega + \varepsilon^2 \omega_2 + \varepsilon^3 \omega_3 + \cdots$$

is associative iff the *Maurer–Cartan* equation holds:

$$[m_\varepsilon, m_\varepsilon]_G = 0 \iff \begin{cases} \delta\omega = 0, & O(\varepsilon), \\ \delta\omega_2 = -\frac{1}{2}[\omega, \omega]_G, & O(\varepsilon^2), \\ \delta\omega_3 = -[\omega, \omega_2]_G, & O(\varepsilon^3), \\ \vdots \end{cases} \quad (72)$$

Theorem 3 (Second-order obstruction). *Let $\omega \in Z^2(\mathcal{A}, \mathcal{A})$ be local and covariant. The class of the Gerstenhaber square $[\omega, \omega]_G$ in $HH^3(\mathcal{A}, \mathcal{A})$ is the only obstruction to lifting the first-order deformation to the second order: there exists ω_2 solving the $O(\varepsilon^2)$ equation in (72) iff $[\omega, \omega]_G = 0$ in HH^3 ; moreover, any two solutions differ by a Hochschild coboundary.*

Sketch. The $O(\varepsilon^2)$ equation is $\delta\omega_2 = -\frac{1}{2}[\omega, \omega]_G$. The right-hand side is δ -closed by graded Jacobi and $\delta\omega = 0$. Solvability requires it to be δ -exact; the obstruction lives in HH^3 . Uniqueness mod B^2 is standard deformation theory à la Gerstenhaber. $\square$

Proposition 12 (Locality through second order). *If ω is a local bidifferential operator supported on the diagonals and $[\omega, \omega]_G$ is δ -exact through a local 2-cochain (i.e., there exists ω_2 local solving the second equation in (72)), then microcausality and the $*$ -structure persist as $O(\varepsilon^2)$ under the same hypotheses as in Section 4.*

Sketch. Locality of ω_2 follows from the microlocal extension arguments used for ω ; the $*$ -compatibility is inherited if ω_2 is built from Z_2 preserving the causal unitarity normalization. The causal support analysis extends because only partial diagonals contribute at this order.

For a precise set of hypotheses under which such a local ω_2 can be chosen, see Section 8.3. $\square$

8.3. Hypotheses for the Existence of a Local ω_2

Proposition 12 established in sketch form that locality and the $*$ -structure persist for $O(\varepsilon^2)$ provided a local solution ω_2 to the Maurer–Cartan equation exists. We now record explicit hypotheses under which such a local ω_2 can indeed be chosen.

Definition 1 (Local exactness of the obstruction). *Let $\omega \in C_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$ with $\delta\omega = 0$. We say that the Gerstenhaber square $[\omega, \omega]_G$ is locally δ -exact if there exists a 2-cochain $\chi \in C_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$ such that*

$$[\omega, \omega]_G = \delta\chi \quad \text{in } C_{\text{loc}}^3(\mathcal{A}, \mathcal{A}).$$

Hypothesis 1 (Existence condition). *Assume that*

- (H1) ω is a local Hochschild 2 cocycle supported on diagonals, covariant under embeddings, and a biderivation modulo local contacts (as in Section 3.6);
- (H2) The obstruction $[\omega, \omega]_G$ is locally δ -exact in the sense above, i.e., $[\omega, \omega]_G = \delta\chi$ with χ itself local and covariant.

Proposition 13 (Local existence of ω_2). *Under Hypothesis 1, there exists a local, covariant cochain $\omega_2 \in C_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$ such that*

$$\delta\omega_2 = -\frac{1}{2}[\omega, \omega]_G,$$

so that $m + \varepsilon\omega + \varepsilon^2\omega_2$ defines an associative product to order ε^2 .

Proof. By (H2) there exists $\chi \in C_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$ with $[\omega, \omega]_G = \delta\chi$. Set $\omega_2 := -\frac{1}{2}\chi$. Then $\delta\omega_2 = -\frac{1}{2}\delta\chi = -\frac{1}{2}[\omega, \omega]_G$. Since χ is assumed local and covariant, so is ω_2 . Associativity then follows from the standard Maurer–Cartan expansion to order ε^2 . $\square$

Diagrammatic summary.

The content of Proposition 13 is summarized in Figure 3: the second-order obstruction lives in HH_{loc}^3 and a local lift ω_2 exists precisely when this class is δ -exact *within* the local subcomplex. For clarity, Figure 4 rephrases the same statement at the level of *cochain quotients*, showing the induced obstruction map on $HH_{\text{loc}}^\bullet$.

$$\begin{array}{ccc}
 HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) & \xrightarrow{\text{Obs} := \frac{1}{2}[\cdot, \cdot]_G} & HH_{\text{loc}}^3(\mathcal{A}, \mathcal{A}) \\
 \downarrow \text{lift to } \omega_2 & & \uparrow \delta \\
 \boxed{\omega_2 \in C_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) : \delta\omega_2 = -\frac{1}{2}[\omega, \omega]_G} & &
 \end{array}$$

Obstruction map and local lifting. The second-order obstruction is the class of $\frac{1}{2}[\omega, \omega]_G$ in HH_{loc}^3 . A lift to order ε^2 exists *locally* iff this class is δ -exact *within the local subcomplex*, i.e., $[\omega, \omega]_G = \delta\chi$ with $\chi \in C_{\text{loc}}^2$; then $\omega_2 = -\frac{1}{2}\chi$ is local and covariant.

Figure 3. Diagrammatic summary of the obstruction map $HH_{\text{loc}}^2 \rightarrow HH_{\text{loc}}^3$ and the local exactness condition for lifting. See also Figure 4 for the induced map on local Hochschild cohomology.

$$\begin{array}{ccc}
 C_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) & \xrightarrow{\frac{1}{2}[\cdot, \cdot]_G} & C_{\text{loc}}^3(\mathcal{A}, \mathcal{A}) \\
 \downarrow \text{mod } B_{\text{loc}}^2 & & \downarrow \text{mod } B_{\text{loc}}^3 \\
 HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) & \xrightarrow{\text{Obs}} & HH_{\text{loc}}^3(\mathcal{A}, \mathcal{A})
 \end{array}$$

Quotient view. The obstruction is the image of $[\omega]$ under the induced map on local Hochschild cohomology. Local solvability requires that $\text{Obs}([\omega]) = 0$, i.e., exactness *inside* the local subcomplex.

Figure 4. Obstruction as the induced map on $HH_{\text{loc}}^\bullet$. Compare with Figure 3 for the explicit lifting picture at the cochain level.

Remark 5. Hypotheses (H1)–(H2) isolate the precise condition required for local solvability: it is not enough that $[\omega, \omega]_G$ vanish in the full Hochschild cohomology, but its trivialization must be accomplished *within the local subcomplex* $C_{\text{loc}}^\bullet$. This ensures that the resulting ω_2 preserves all structural axioms (locality, covariance, scaling bounds, $*$ —compatibility). In general interacting models, establishing (H2) is a nontrivial problem; verifying it case by case (e.g., for free fields and low-order Wick monomials) is the natural first step toward higher-order consistency.

Remark 6 (Renormalization scheme dependence and $HH^\bullet$). At second and higher orders, different choices of local ω_k related by δ -exact shifts correspond to finite renormalizations and do not alter the cohomology class. The tower (72) can be read as a nested sequence of cohomological equations whose solvability determines the formal moduli of associative structures near m ; see also the L_∞ interpretation in Section 8.5.

8.4. Net-Level and Wedge-Local Deformations

The present deformation is formulated at the level of the algebra of microcausal functionals. It is natural to ask for *net-level* deformations $O \mapsto \mathcal{A}_\varepsilon(O)$ preserving isotony and some notion of locality. Two well-developed paradigms are (i) *Rieffel deformations* of C^* -algebras under $\mathbb{R}^d$ -actions and (ii) *warped convolutions* producing wedge-local QFTs.

Definition 2 (Composed deformation). Let $\mathcal{A}(O)$ be a Haag–Kastler net, α_t a strongly continuous action of $\mathbb{R}^d$ by automorphisms, and $\times_\Theta$ is the Rieffel product for a skew form Θ . Define the composed product

$$A \bullet B := (A * B) \times_\Theta 1 \quad \text{on } \mathcal{A}(O),$$

i.e., first apply our local deformation $*$ (scheme change) and then the global Rieffel deformation.

Proposition 14 (First-order additivity). To first order in ε and Θ (small parameters), the composed product is associative, and its first-order cocycle is the sum of the cocycles:

$$A \bullet B = AB + \varepsilon \omega(A, B) + \frac{i}{2} \Theta^{\mu\nu} (\partial_\mu A)(\partial_\nu B) + O(\varepsilon^2, \Theta^2, \varepsilon\Theta).$$

In particular, wedge locality inherited from the warped convolution survives for $O(\varepsilon)$.

Sketch. Both deformations are implemented by formal automorphisms (Stückelberg–Petermann for $*$ and oscillatory integrals for $\times_\Theta$). Expanding to the first order in each parameter, the cross-term $O(\varepsilon\Theta)$ is of second order and can be dropped; associativity follows from the separate associativity to first order. $\square$

Remark 7 (Drinfel’d twists and quantum symmetries). Alternatively, one may implement a twist of the symmetry Hopf algebra $\mathcal{U}(\mathfrak{p})$ (Poincaré) by a Drinfel’d twist $\mathcal{F}$ and let $\mathcal{F}$ act on the net [59,60]. Our deformation is local and renormalization-driven rather than symmetry-driven; nevertheless, at first order the two can be superposed as in Proposition 14. Twisted Poincaré-invariant models provide explicit wedge-local realizations [61–63].

8.5. Operadic and Factorization-Algebra Lifts

The pAQFT time-ordered product endows local observables with a factorization structure; one can attempt to lift the deformation to the *homotopy-coherent* level. Let O denote the operan that governs the algebras of observables (e.g., E_d -type or a factorization algebra in the sense of Costello–Gwilliam).

Proposition 15 (Deformation complex and L_∞ Maurer–Cartan). Let $\text{Def}(\mathcal{A})$ be the deformation complex of the O -algebra of observables (a dg Lie algebra). Then the associative deformation problem (72) is controlled by an L_∞ Maurer–Cartan element in $\text{Def}(\mathcal{A})$ whose linear term is the Hochschild cocycle ω ; the higher brackets encode the obstructions in HH^3 and beyond.

Sketch. By general operadic deformation theory, deformations of an O -algebra are governed by the convolution dg Lie algebra $\text{Hom}(O^\vee, \text{End}_{\mathcal{A}})$; the linearization recovers Hochschild for associative O and Harrison for commutative O [64,65]. The pAQFT factorization structure identifies the relevant O (locally, E_d -type), placing ω as the first Taylor component of an L_∞ solution. $\square$

Remark 8 (Factorization homology and higher Hochschild). Higher Hochschild (Pirashvili) cohomology appears naturally for E_d -algebras; the locality of ω indicates that the relevant classes live in topologically supported variants of Hochschild adapted to factorization algebras [66–69].

8.6. Hopf-Algebraic Renormalization and the Curvature Class

The Connes–Kreimer Hopf algebra $\mathcal{H}$ of Feynman graphs organizes renormalization through the Birkhoff decomposition of characters with values in a Rota–Baxter algebra [70]. Let $\phi_\mu : \mathcal{H} \rightarrow \mathcal{A}$ be the character induced by a renormalization scheme at scale μ and $\gamma = \phi_-^{-1} \star \phi_+$ the counterterm-renormalized factorization.

Proposition 16 (Differentiating the Birkhoff factorization). *The infinitesimal variation of the character under a local change of scheme induces a Hochschild 1-cochain η on $\mathcal{A}$; its Hochschild coboundary $\delta\eta$ reproduces the identity component of the local contact terms that define ω . Consequently, the curvature class $[\omega]$ can be extracted from the logarithmic derivative of the counterterm character modulo exact pieces.*

Sketch. Variations of $\phi_{\pm}$ correspond to variations of the projection onto polar/regular parts in the Rota–Baxter algebra; differentiating the convolution $\phi_{-}^{-1} \star \phi_{+}$ yields a derivation on amplitudes whose failure to be an inner derivation is captured by $\delta\eta$ [71]. Locality ensures that only contact contributions survive, matching the identity terms in ω . $\square$

8.7. Gravitational and Cosmological Sectors

We outline a consistent decoupling of gravitational sectors and state a no-go for “single-parameter universality” across matter and gravity.

Two-parameter deformation.

Introduce a distinct gravitational deformation parameter $\varepsilon_{\text{grav}}$ so that the total product reads

$$A \star B = AB + \varepsilon \omega_{\text{matt}}(A, B) + \varepsilon_{\text{grav}} \omega_{\text{grav}}(A, B) + \cdots,$$

where ω_{grav} acts on composite operators involving the metric and curvature and is constrained by diffeomorphism covariance and scaling degree (dimensionful couplings).

Proposition 17 (Dimensional obstruction to universality). *No choice of a single dimensionless parameter ε can produce scheme-independent, dimensionally consistent, nontrivial first-order correlations between (i) matter dipole observables (dimensionless a_{ℓ} , dimension M^{-1} EDMs) and (ii) purely gravitational composite quantities with a nonzero mass dimension (e.g., $\langle T^{\mu}_{\mu} \rangle$, curvature invariants) across all backgrounds.*

Sketch. Matter dipoles arise from operators of canonical dimension 5 (after EWSB) whose Wilson coefficients scale as $1/\Lambda^2$. Gravitational composites involve curvature tensors with mass dimension 2 or higher. A single dimensionless ε multiplying a *local* ω cannot convert the dimensionless $(g-2)$ into a scale-independent prediction for, e.g., $\langle R \rangle$ or Λ_{eff} without introducing explicit scales beyond Λ . Since the scheme dependence cancels only within sectors of the same engineering dimension, cross-correlating sectors would either violate dimensional analysis or reintroduce scheme-dependent scales. Therefore, any nontrivial, universal correlation requires at least one additional dimensionful (or dimensionless) parameter controlling the gravitational deformation. $\square$

Decoupling and backreaction.

To the first order, one may consistently set $\varepsilon_{\text{grav}} = 0$ when computing matter observables on fixed backgrounds; backreaction appears through $\mathcal{O}(\varepsilon_{\text{grav}})$ corrections to the expectation value of $T_{\mu\nu}$ and enters the semiclassical Einstein equations via local covariant counterterms. The anomaly map of Section 6 extends: Weyl anomalies lie in the image of $\Phi([\omega_{\text{grav}}])$ but are controlled independently of $[\omega_{\text{matt}}]$. A compact summary of the main consistency conditions and extension avenues beyond first order is provided in Table 2.

Table 2. Checklist of consistency conditions and extension avenues beyond first order.

Aspect	Status/Condition
Associativity at $O(\varepsilon)$	$\delta\omega = 0$ (Hochschild 2-cocycle)
Associativity at $O(\varepsilon^2)$	$[[\omega, \omega]_G] = 0$ in HH^3 ; choose local ω_2
*-structure and microcausality	Local Z_k preserving causal unitarity $\Rightarrow$ (15) to each order
WTIs/ST identities	Biderivation modulo contact terms; contact terms renormalized away at first order
Net/wedge locality	Composed deformations additive at first order (Proposition 14)
Operadic lift	L_∞ Maurer–Cartan in deformation complex (Proposition 15)
Gravity/cosmology	Separate parameter $\varepsilon_{\text{grav}}$; no single-parameter universality (Proposition 17)

8.8. Outlook of This Section

The first-order “algebraic curvature” picture is robust under several refinements: it nests coherently in the full Maurer–Cartan tower, it is composed with net-level (wedge-local) deformations, it admits an L_∞ lift aligned with factorization algebra technology, and it decouples cleanly from gravity at leading order. The main technical challenges for future work are (i) identifying sufficient conditions for the *local* solvability of the HH^3 obstruction in interacting models; (ii) constructing explicit higher-order counterterms Z_k that preserve the biderivation property relative to BRST; (iii) encoding ω in the Hopf-algebraic renormalization group in a way that makes contact with global geometric invariants beyond one loop.

9. Outlook and Conceptual Perspectives

This section synthesizes the picture that has emerged and places it within broader mathematical and physical frameworks. At its core, the present work argues that changes of renormalization prescription in perturbative AQFT define a *flatness problem* for an algebraic connection on the “bundle of products,” with a first-order curvature measured by a local Hochschild class $[\omega] \in HH^2_{\text{loc}}(\mathcal{A}, \mathcal{A})$. This *algebraic curvature* is functorially related to standard anomaly cohomology and feeds universal, low-energy correlations among transverse form factors. We now develop geometric, categorical, and information-theoretic viewpoints that motivate further developments and suggest concrete tests.

9.1. Geometry on the Renormalization Torsor

Let $\mathfrak{S}$ be the (infinite-dimensional) manifold of renormalization prescriptions T obeying the pAQFT axioms, factored by natural equivalences (local covariance, microlocal spectrum, etc.). The Stückelberg–Petermann group SP acts freely and transitively on each connected component of $\mathfrak{S}$, making $\mathfrak{S}$ a torsor for SP . Let $\mathcal{M}$ denote the space of associative products on $\mathcal{A}$ compatible with the axioms (again modulo natural equivalences). Transport of the structure $T \mapsto *_T$ yields a principal SP -bundle $\pi : \mathcal{P} \rightarrow \mathcal{M}$; at first order near T we may model SP by its Lie algebra of local 1-cochains.

Definition 3 (Algebraic renormalization connection). *A choice of splitting the short exact sequence*

$$0 \longrightarrow \text{Inn}_{\text{loc}}(\mathcal{A}) \longrightarrow \text{Der}_{\text{loc}}(\mathcal{A}) \longrightarrow \mathfrak{sp}_{\text{loc}} \longrightarrow 0$$

(where Der_{loc} and Inn_{loc} are local derivations and inner derivations, and $\mathfrak{sp}_{\text{loc}}$ is the Lie algebra of local Stückelberg–Petermann transformations) defines an Ehresmann connection A on $\pi : \mathcal{P} \rightarrow \mathcal{M}$, with values in local derivations of $\mathcal{A}$.

Infinitesimally, a horizontal lift of a tangent vector $\dot{m} \in T_m \mathcal{M}$ picks a local Z_1 ; the corresponding first-order product deformation is

$$m \mapsto m + \varepsilon \omega, \quad \omega = \delta Z_1,$$

computed in the *full* Hochschild complex. The fact that we work in the *local* subcomplex is crucial: while δZ_1 is a Hochschild coboundary globally, its class $[\omega] \in HH_{\text{loc}}^2$ need *not* vanish when we restrict to cochains compatible with locality, covariance, and scaling (Sections 3 and 4).

Proposition 18 (Curvature of the renormalization connection). *Let $F = dA + A \wedge A$ be the curvature of A . Evaluated on two tangent directions $\dot{m}_1, \dot{m}_2$ and projected to the local Hochschild complex, $F(\dot{m}_1, \dot{m}_2)$ represents a class*

$$[F] \in HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A})$$

which coincides (to the first order) with the algebraic curvature class $[\omega]$ constructed from the corresponding pair of local lifts $Z_1^{(i)}$. In particular, $[\omega] = 0$ if and only if there exists a local gauge of π in which the product is constant along the given 2-plane in $T_m \mathcal{M}$.

Sketch. Given local lifts $Z_1^{(i)}$ along $\dot{m}_i$, the commutator $[Z_1^{(1)}, Z_1^{(2)}]$, and the difference of mixed partials produce a 2-cochain that equals the curvature of A ; projecting to local cochains and using δ -functoriality identifies the class with that of the induced ω . $\square$

This viewpoint clarifies the slogan “*renormalization ambiguities carry curvature*”: $[\omega]$ is the curvature of an algebra-valued connection on the bundle of associative structures. Flatness corresponds to the trivialization of scheme dependence by local inner automorphisms; nonzero curvature indicates a genuine obstruction, captured cohomologically.

9.2. Index Theory, Descent, and Anomaly Inflow

Section 6 constructed a homomorphism

$$\Phi : HH_{\text{loc}}^2(\mathcal{A}, \mathcal{A}) \longrightarrow H^{1,d}(s|d),$$

sending $[\omega]$ to a BRST anomaly class via the anomalous Master Ward Identity. From the geometric standpoint, Φ factors through the *symbol* of the local bidifferential operator ω and then through the Stora–Zumino descent associated with a characteristic class I_{d+2} :

$$HH_{\text{loc}}^2 \xrightarrow{\sigma} \Omega_{\text{inv}}^{d+2}(BG) \xrightarrow{\text{descent}} H^{1,d}(s|d).$$

Here, $\Omega_{\text{inv}}^{d+2}(BG)$ denotes invariant polynomials on the gauge curvature (and Riemann curvature for gravitational anomalies). This makes explicit contact with the local index theorem: I_{d+2} is the index density for a suitable family of Dirac-type operators [72,73], and the descent constructs the consistent anomaly. The inflow mechanism [74] then realizes $a^{1,d}$ as the boundary variation of a $(d+1)$ -dimensional invertible theory with Chern–Simons action built from I_{d+2} .

Proposition 19 (Compatibility with anomaly inflow). *Let I_{d+2} be the characteristic form obtained from $\sigma(\omega)$. The boundary variation of the $(d+1)$ -dimensional Chern–Simons functional with transgression form $CS_{d+1}(A)$ associated with I_{d+2} reproduces $\Phi([\omega])$ up to s - and d -exact terms.*

Sketch. $dCS_{d+1}(A) = I_{d+2}(F)$; a gauge variation $\delta_\alpha CS_{d+1}$ is exact and integrates over the boundary to the Wess–Zumino consistent anomaly determined by the descent of I_{d+2} [75]. The symbol map σ identifies the principal part of ω with $I_{d+2}(F)$, ensuring identical cohomology classes. $\square$

9.3. Anomalies as Invertible Field Theories and Categorical Lifts

Modern treatments view anomalies as *invertible* $(d+1)$ -dimensional field theories classified by generalized cohomology groups [76,77]. From this perspective, the functor Φ can be interpreted as a boundary map from local Hochschild data to the class of the bulk invertible theory. This suggests a categorical refinement: upgrading $(\mathcal{A}, *)$ from an associative algebra to a factorization algebra (or E_d -algebra) on spacetime and promoting ω to the first Taylor coefficient of an L_∞ Maurer–Cartan element in the deformation complex (Section 8.5). The bulk invertible theory then appears as the “anomaly object” implementing the failure of strict functoriality; its cobordism class is determined by $\sigma(\omega)$.

Remark 9 (Global anomalies). *The discussion thus far concerns local anomalies. Global anomalies (e.g., Witten’s $SU(2)$ anomaly [78]) fit naturally into the invertible-TQFT picture via torsion classes in the relevant cobordism groups [76]. While $[\omega]$ detects local curvature, a full classification requires supplementing Φ with topological data; this lies beyond the local-Hochschild scope but is compatible with our framework when extended to the global setting.*

9.4. Information Geometry, Modular Theory, and “Quantum Curvature”

Beyond symmetry/anomaly, the curvature encoded by $[\omega]$ resonates with *quantum information geometry*. Consider a smooth family of faithful states $\{\varphi_\lambda\}$ on a fixed local algebra $\mathcal{A}(O)$. The Uhlmann connection on the bundle of purifications [79] defines a quantum parallel transport whose curvature captures mixed-state geometric phases and is closely related to monotone quantum metrics [80]. On the algebraic side, Tomita–Takesaki modular theory associates the modular operator Δ_λ and modular flow with $(\mathcal{A}(O), \varphi_\lambda)$; variations in λ define a “modular connection” and associated susceptibilities (relative entropy Hessians) [81,82].

Proposition 20 (First-order compatibility with modular response). *Let A be the renormalization connection of Section 9.1. For a family φ_λ of KMS states smooth in λ , the linear response of modular generators under a horizontal shift of A is a local derivation, and its commutator across two horizontal directions agrees, to the first order, with the algebraic curvature class $[\omega]$ applied to modular Hamiltonians, modulo local counterterms.*

Sketch. Horizontal shifts correspond to local reparametrizations Z_1 ; modular generators change by derivations implementing Z_1 on local observables. The commutator of two such changes produces a bidifferential operator supported on diagonals, i.e., a representative of ω acting on the modular Hamiltonian density. Local counterterms arise from the ambiguity in choosing Wick polynomials and vanish in the class $[\omega]$. $\square$

This points to a unifying theme: *algebraic curvature* governs not only anomaly classes but also aspects of quantum geometric response, suggesting links to c -/ a -functions (metric on theory space) and to constraints from monotonicity (e.g., Zamolodchikov’s metric in $d=2$ [83]).

9.5. Beyond Perturbation Theory and Low Dimensions

Nonperturbative constructions in low dimensions provide testing grounds. In $1+1$ dimensions, integrable models and factorizing S -matrices admit rigorous algebraic realizations; wedge-local fields can be constructed directly and deformed while preserving

modular nuclearity. Embedding our first-order local deformation into such models—or using form-factor bootstrap to compute the induced $F_2(0)$ shifts—would test robustness beyond perturbation theory. In higher dimensions, constructive progress remains challenging, but wedge-local deformations and operator-algebraic methods offer avenues for partial results (Haag–Kastler nets built from polarization-free generators, etc.).

9.6. Programmatic Experimental Targets

From Section 7, the most incisive predictions are as follows:

- **EDM- $g - 2$ phase relation** (63): $d_\ell/\Delta a_\ell = (e/2m_\ell) \tan \phi_\ell$, robust under running and matching.
- **Flavor-scaling laws** (Table 1): distinguish Texture U (universal) from Texture Y (Yukawa-aligned) by comparing Δa_e to Δa_μ and d_e to d_μ .
- **Parity-violating observables**: anapole contributions controlled by axial pieces of ω can be probed in polarized Møller scattering and atomic parity violation; correlations with dipoles test the tensor decomposition of ω .

A near-term strategy is to calibrate ε in the muon channel and propagate to electron observables; improved bounds on d_e and prospective muon EDM searches then constrain ϕ_ℓ and thus the CP-odd sector of ω .

9.7. Open Problems and Roadmap

1. **Higher orders**: Prove the existence of *local* solutions to the obstruction equations in (72) for interacting models and classify HH_{loc}^3 under pAQFT axioms.
2. **BRST compatibility**: Construct Z_k preserving biderivation properties relative to the BRST differential beyond the first order; relate to the renormalized BV Quantum Master Equation.
3. **Bulk refinement**: Define a functor from HH_{loc}^2 to the spectrum classifying invertible $(d+1)$ -dimensional field theories, making Φ fully functorial (beyond the local symbol).
4. **Modular geometry**: Develop a rigorous link between $[\omega]$ and modular Berry/Uhlmann curvature for families of local KMS states; explore monotonicity constraints.
5. **Nonperturbative tests**: Implement first-order deformations in wedge-local/integrable models and compute observable shifts.
6. **Gravitational decoupling**: Construct ω_{grav} and analyze Weyl anomalies purely from algebraic curvature, including conditions for scheme independence in curved backgrounds.

9.8. Concluding Perspective

Renormalization ambiguities are often regarded as a technical nuisance; here, they acquire a *geometric identity*. The local Hochschild class $[\omega]$ is an algebraic curvature measured by the failure of scheme changes to be strictly inner when constrained by locality and covariance. It not only reproduces the familiar anomaly cohomology but also imprints universal signatures on low-energy amplitudes. The resulting picture—a connection on the space of products, with curvature feeding both anomalies and dipole correlations—suggests that *geometry can and should organize the interface between microscopic definitions and macroscopic observables* in quantum field theory.

For additional background and standard sources relevant to deformation/renormalization, Hopf-algebraic renormalization, and EFT phenomenology used as context in this work, see [84–109].

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Appendix A. Microlocal Preliminaries and Explicit Constructions of ω

Appendix A.1. Wavefront Sets, Scaling Degree, and Microcausal Functionals

Let M be a globally hyperbolic spacetime with metric g and volume form dvol_g . For $u \in \mathcal{D}'(M^n)$, its wavefront set $\text{WF}(u) \subset T^*(M^n) \setminus \{0\}$ controls the singular directions of u [110,111]. A distribution $u(x, y)$ is *Hadamard* if $\text{WF}(u)$ is contained in the set of pairs of lightlike covectors with future-directed behavior; the *Hadamard two-point function* ω_2 and the *Feynman parametrix* H_F obey this criterion.

The *scaling degree* of $t \in \mathcal{D}'(\mathbb{R}^{Nd} \setminus \{0\})$ at 0 is

$$\text{sd}_0(t) := \inf \left\{ \delta \in \mathbb{R} : \lim_{\lambda \downarrow 0} \lambda^\delta t(\lambda x) = 0 \text{ in } \mathcal{D}'(\mathbb{R}^{Nd} \setminus \{0\}) \right\}.$$

If $\text{sd}_0(t) < \infty$, there exist extensions $\bar{t} \in \mathcal{D}'(\mathbb{R}^{Nd})$ with $\text{sd}_0(\bar{t}) = \text{sd}_0(t)$; non-uniqueness occurs when $\omega := \text{sd}_0(t) - Nd \geq 0$, with ambiguity spanned by $\partial^\alpha \delta$ for $|\alpha| \leq \omega$.

A *microcausal functional* F on the space of classical fields has functional derivatives $F^{(n)}(\varphi) \in \mathcal{D}'(M^n)$ with

$$\text{WF}(F^{(n)}(\varphi)) \cap (\bar{V}_+^n \cup \bar{V}_-^n) = \emptyset,$$

ensuring well-defined star/time-ordered products with Hadamard two-point functions [112].

Appendix A.2. Free-Field Star Product, Time-Ordered Products, and EG Extension

For a Hadamard ω_2 , define the bidifferential operator

$$\Gamma_{\omega_2} = \frac{1}{2} \int_{M \times M} \omega_2(x, y) \frac{\delta}{\delta \varphi(x)} \otimes \frac{\delta}{\delta \varphi(y)} \text{dvol}_g(x) \text{dvol}_g(y),$$

and the star product $F \star G := \mu \circ \exp(\hbar \Gamma_{\omega_2})(F \otimes G)$, with μ pointwise multiplication. Time-ordered products are defined similarly with the Feynman parametrix H_F :

$$T = \exp(\hbar \Gamma_{H_F}), \quad \Gamma_{H_F} = \frac{1}{2} \int H_F(x, y) \frac{\delta}{\delta \varphi(x)} \otimes \frac{\delta}{\delta \varphi(y)}.$$

Products of parametrices produce distributions singular on diagonals; extension to diagonals is performed by the Epstein–Glaser (EG) recursion governed by scaling degree. Ambiguities are local and of finite order.

Appendix A.3. Scalar Model: $:\phi^2:$ and $:\phi^3:$ Examples (Full Details)

Let $:\phi^k: (f) := \int :\phi(x)^k: f(x) \text{dvol}_g(x)$ with respect to ω_2 .

The $:\phi^2: \times :\phi^2:$ calculation.

Wick's theorem yields, for $x \neq y$,

$$T(:\phi^2: (x), :\phi^2: (y)) = :\phi^2(x)\phi^2(y): + 4 H_F(x, y) :\phi(x)\phi(y): + 2 H_F(x, y)^2. \quad (\text{A1})$$

The term H_F^2 has $\text{sd}_0(H_F^2) = 4$ in $d = 4$, so $\omega = 0$ and the extension is unique up to a linear combination $\sum_{|\alpha| \leq 2} a_\alpha \partial^\alpha \delta(x - y)$. Thus

$$\Delta T_2(:\phi^2: (f), :\phi^2: (g)) = 2 \sum_{|\alpha| \leq 2} \Delta a_\alpha \int \partial^\alpha (fg)(x) \mathbf{1} \text{dvol}_g(x) \quad (\text{A2})$$

and, transporting via $Z = \text{id} + \varepsilon Z_1 + \dots$,

$$\omega(\colon\phi^2\colon(f), \colon\phi^2\colon(g)) = 2 \sum_{|\alpha| \leq 2} \Delta a_\alpha \int \partial^\alpha(fg)(x) \mathbf{1} \, \text{dvol}_g(x). \quad (\text{A3})$$

The $\colon\phi^3\colon \times \colon\phi\colon$ calculation.

For $x \neq y$,

$$T(\colon\phi^3\colon(x), \colon\phi\colon(y)) = \colon\phi^3(x)\phi(y)\colon + 3H_F(x, y) \colon\phi^2(x)\colon.$$

$\text{sd}_0(H_F) = 2$ in $d=4$, hence $\omega = -2 + 4 - 2 = 0$ for the product $H_F\delta$, and the extension entails a single derivative order. A change in extension gives

$$\omega(\colon\phi^3\colon(f), \colon\phi\colon(g)) = 3 \sum_{|\alpha| \leq 1} \Delta b_\alpha \int \partial^\alpha(fg)(x) \colon\phi^2(x)\colon \, \text{dvol}_g(x). \quad (\text{A4})$$

Cocycle check.

Define $\omega(A, B)$ by Z_1 via $\omega(A, B) = Z_1(AB) - Z_1(A)B - AZ_1(B)$ on Wick monomials. Then $\delta\omega = 0$ follows from the identity $\delta = [m, \cdot]_G$ and the associativity of the transported product to $O(\varepsilon)$; explicitly, for three monomials F_i ,

$$\delta\omega(F_1, F_2, F_3) = F_1\omega(F_2, F_3) - \omega(F_1F_2, F_3) + \omega(F_1, F_2F_3) - \omega(F_1, F_2)F_3 = 0.$$

Each term is supported on (unions of) diagonals; away from diagonals, the Leibniz rule holds, and the alternating sum vanishes, while on diagonals, the EG extension preserves the Hochschild identity by construction.

Appendix A.4. Dirac Fields: Current–Field and Current–Current Products

Let S_F be the Dirac Feynman kernel. For $J^\mu(x) = \colon\bar{\psi}\gamma^\mu\psi\colon(x)$ and $\psi(y)$,

$$T(J^\mu(x), \psi(y)) = \colon\bar{\psi}(x)\gamma^\mu\psi(x)\psi(y)\colon - \gamma^\mu S_F(x, y) \psi(y). \quad (\text{A5})$$

Extension at $x = y$ modifies the second term by local contributions:

$$\Delta T(J^\mu(f), \psi(g)) = \sum_{|\alpha| \leq 1} c^\mu_\alpha \int \partial^\alpha(fg)(x) \psi(x) \, \text{dvol}_g(x), \quad (\text{A6})$$

so

$$\omega(J^\mu(f), \psi(g)) = \sum_{|\alpha| \leq 1} c^\mu_\alpha \int \partial^\alpha(fg) \psi.$$

For $J^\mu - J^\nu$,

$$T(J^\mu(x), J^\nu(y)) = \colon J^\mu(x)J^\nu(y)\colon + \text{tr}(\gamma^\mu S_F(x, y)\gamma^\nu S_F(y, x)),$$

and extension yields

$$\omega(J^\mu(f), J^\nu(g)) = \sum_{|\alpha| \leq 2} d^{\mu\nu}_\alpha \int \partial^\alpha(fg)(x) \mathbf{1} \, \text{dvol}_g(x).$$

The $\ast$ -compatibility $\omega(A, B)^\dagger = \omega(B^\dagger, A^\dagger)$ holds because Z_1 intertwines adjunction (causal unitarity normalization).

Appendix A.5. Curvature Dependence via Hadamard Coefficients

On a curved background,

$$H_F(x, y) = \frac{U(x, y)}{\sigma_+(x, y)} + V(x, y) \log \sigma_+(x, y) + W(x, y),$$

with U, V, W smooth bidensities determined by transport equations; products H_F^2 and Dirac traces admit point-split expansions whose extensions differ by curvature polynomials multiplying derivatives of $\delta(x - y)$. Therefore, the coefficients in ω become local polynomials in $R, R_{\mu\nu}$ and their derivatives with scaling fixed by sd_0 .

Appendix B. Associativity to Second Order and Obstructions in HH^3

Appendix B.1. Gerstenhaber Calculus and the Maurer–Cartan Tower

Let m be the original product and $m_\varepsilon = m + \varepsilon\omega + \varepsilon^2\omega_2 + \dots$. Associativity is equivalent to $[m_\varepsilon, m_\varepsilon]_G = 0$, which expands to

$$O(\varepsilon) : \quad \delta\omega = 0, \tag{A7}$$

$$O(\varepsilon^2) : \quad \delta\omega_2 = -\frac{1}{2}[\omega, \omega]_G, \tag{A8}$$

$$O(\varepsilon^3) : \quad \delta\omega_3 = -[\omega, \omega_2]_G, \text{ etc.} \tag{A9}$$

with $\delta\phi = [m, \phi]_G$.

Appendix B.2. Second-Order Obstruction and Locality

Theorem A1. Let $\omega \in Z^2(\mathcal{A}, \mathcal{A})$ be local and covariant. A necessary and sufficient condition for the existence of a local ω_2 solving $\delta\omega_2 = -\frac{1}{2}[\omega, \omega]_G$ is that the class of $[\omega, \omega]_G$ vanishes in $HH_{\text{loc}}^3(\mathcal{A}, \mathcal{A})$.

Proof. Necessity is immediate: if ω_2 exists locally, $[\omega, \omega]_G$ is δ -exact in the local subcomplex. For sufficiency, assume $[\omega, \omega]_G = \delta\chi$ with χ a local 2-cochain; then $\omega_2 := -\frac{1}{2}\chi$ solves the equation and is local. Local covariance follows from the naturality of the EG extension and of Z_2 . $\square$

Appendix B.3. Preservation of $\ast$ -Structure and Microcausality

If Z_2 is chosen to preserve causal unitarity and field independence, then $(A \ast B)^\dagger = B^\dagger \ast A^\dagger$ holds to $O(\varepsilon^2)$. Microcausality persists because ω_2 is supported on diagonals; commutators vanish at spacelike separation up to contact terms supported on measure-zero sets (which integrate to zero against test functions with separated supports).

Appendix B.4. Biderivations Modulo Contact Terms and WTIs

Assume ω obeys the biderivation identities modulo local contacts:

$$\omega(A, BC) = \omega(A, B)C + B\omega(A, C) + \text{l.c.t.},$$

and similarly in the first argument. Then the first-order current algebra deforms by local terms only; the WTI is preserved to $O(\varepsilon)$ after normalization. At $O(\varepsilon^2)$ one needs compatibility between ω_2 and BRST variations; this amounts to the $O(\varepsilon^2)$ MWI with anomaly controlled by the class of $[\omega, \omega]_G$.

Appendix C. Anomalous Master Ward Identity in pAQFT

Appendix C.1. BV Setup and Renormalized Time-Ordered Products

Introduce fields Φ^A , antifields $\Phi_A^\dagger$, and the BV antibracket $(\cdot, \cdot)$. Let S be the classical gauge-fixed action with $(S, S) = 0$, and $s := (S, \cdot)$ the BRST differential. Renormalized time-ordered products T_n act on local functionals of $(\Phi, \Phi^\dagger)$ and obey causal factorization, microlocal spectrum, and locality.

Appendix C.2. Quantum Master Equation and the MWI

The renormalized Quantum Master Equation (QME) reads

$$\frac{1}{2}(S, S) + \hbar \Delta S = 0,$$

with Δ the renormalized BV Laplacian. In pAQFT, Δ acts on local functionals and produces local contact terms [113,114]. For a time-ordered exponential $T(e^{\frac{i}{\hbar} F})$, one has the *Master Ward Identity* (MWI)

$$s T(X) = \frac{i}{\hbar} T(sX) + \frac{i}{\hbar} T(\mathcal{A}(X)), \quad \mathcal{A}(X) \text{ local and supported on diagonals.} \quad (\text{A10})$$

The anomaly map $\mathcal{A}$ is a Hochschild 1-cocycle on the time-ordered algebra and satisfies the Wess–Zumino condition $s\mathcal{A} + \mathcal{A}s = -d(\cdots)$ at the level of densities.

Appendix C.3. Effect of the Product Deformation

Let $*$ be the deformed product $AB + \varepsilon\omega(A, B)$, and T_* the deformed time-ordered products (transported by Z). Then

$$s T_*(X) = \frac{i}{\hbar} T_*(sX) + \frac{i}{\hbar} T_*(\mathcal{A}_*(X)), \quad \mathcal{A}_*(X) = \mathcal{A}(X) + \varepsilon \mathcal{A}_\omega(X) + O(\varepsilon^2),$$

with $\mathcal{A}_\omega$ computed from the local contact terms generated by ω (explicit form (49) in the main text).

Proposition A1 (WZ consistency at first order). *If $\delta\omega = 0$ and ω is a biderivation modulo contacts, then*

$$s \mathcal{A}_\omega(X) = -d \mathcal{B}_\omega(X),$$

with $\mathcal{B}_\omega$ local; hence $a_\omega^{1,d}$, the density associated with $\mathcal{A}_\omega$, is a representative of a class in $H^{1,d}(s|d)$.

Proof. Apply s to the MWI for T_* twice and use $s^2 = 0$. The cross-term $s\mathcal{A}_\omega + \mathcal{A}_\omega s$ is a sum of local functionals supported on diagonals; causal factorization organizes them into a total derivative by the standard descent argument. Hochschild closedness of ω supplies the graded Jacobi identity needed for closure. $\square$

Appendix C.4. Current WTIs as a Special Case

For Abelian currents, the MWI reduces to the ordinary WTI of Section 5 when restricted to background gauge transformations; the contact anomaly $\mathcal{A}_\omega$ becomes the divergence of a local polynomial C^μ , which is absorbed by normalization of the conserved current.

Appendix D. SMEFT $\rightarrow$ LEFT Matching and RG Evolution for Dipoles

Appendix D.1. Warsaw Basis, Electroweak Breaking, and Photon Dipole

In the Warsaw basis, the lepton dipoles are

$$Q_{eB}^\ell = (L_\ell \sigma^{\mu\nu} e_{R\ell}) H B_{\mu\nu}, \quad Q_{eW}^\ell = (L_\ell \sigma^{\mu\nu} e_{R\ell}) \tau^I H W_{\mu\nu}^I.$$

The effective Lagrangian at scale Λ contains

$$\mathcal{L} \supset \frac{C_{eB}^\ell}{\Lambda^2} Q_{eB}^\ell + \frac{C_{eW}^\ell}{\Lambda^2} Q_{eW}^\ell + \text{h.c.}$$

After EWSB, $H \rightarrow (0, (v+h)/\sqrt{2})^T$ and $A_\mu = s_\theta W_\mu^3 + c_\theta B_\mu$ with $s_\theta = \sin \theta_W$, $c_\theta = \cos \theta_W$, so the photon dipole coefficient is

$$C_{e\gamma}^\ell(\Lambda) = c_\theta C_{eB}^\ell(\Lambda) - s_\theta C_{eW}^\ell(\Lambda). \quad (\text{A11})$$

Matching onto LEFT yields

$$\mathcal{L}_{\text{LEFT}} \supset \frac{1}{2} \bar{\ell} \sigma^{\mu\nu} (c_\ell + i\tilde{c}_\ell \gamma_5) \ell F_{\mu\nu}, \quad c_\ell + i\tilde{c}_\ell = \frac{v}{\sqrt{2}\Lambda^2} C_{e\gamma}^\ell(\Lambda).$$

Appendix D.2. One-Loop Running in SMEFT (Symbolic Form)

At one loop, the SMEFT dipoles mix among themselves and with other operators (four-fermion, Higgs–lepton). Schematically,

$$16\pi^2 \frac{d}{d \ln \mu} \begin{pmatrix} C_{eB}^\ell \\ C_{eW}^\ell \end{pmatrix} = \underbrace{\begin{pmatrix} \gamma_{BB} & \gamma_{BW} \\ \gamma_{WB} & \gamma_{WW} \end{pmatrix}}_{\text{gauge, Yukawa}} \begin{pmatrix} C_{eB}^\ell \\ C_{eW}^\ell \end{pmatrix} + \underbrace{\sum_j \begin{pmatrix} \gamma_{B \leftarrow j} \\ \gamma_{W \leftarrow j} \end{pmatrix}}_{\text{mixing}} C_j, \quad (\text{A12})$$

where entries are polynomials in $g_1^2, g_2^2, y_\ell^2, \text{Tr}(Y_u^\dagger Y_u)$, etc. Using (A11), one finds the RGE for $C_{e\gamma}^\ell$ by rotation with $(c_\theta, -s_\theta)$.

Appendix D.3. QED Running in LEFT

Below EWSB and lepton thresholds, the QED+QCD running of dipoles is diagonal at leading log:

$$16\pi^2 \frac{d}{d \ln \mu} \begin{pmatrix} c_\ell \\ \tilde{c}_\ell \end{pmatrix} = \begin{pmatrix} \gamma_{cc} & 0 \\ 0 & \gamma_{\tilde{c}\tilde{c}} \end{pmatrix} \begin{pmatrix} c_\ell \\ \tilde{c}_\ell \end{pmatrix} + \sum_j \begin{pmatrix} \gamma_{c \leftarrow j} \\ \gamma_{\tilde{c} \leftarrow j} \end{pmatrix} C_j, \quad (\text{A13})$$

with C_j denoting four-fermion and semileptonic operators that mix into dipoles. To the leading log and neglecting mixing,

$$c_\ell(\mu) \simeq c_\ell(\Lambda) \left[1 + \frac{\gamma_{cc}}{16\pi^2} \ln \frac{\mu}{\Lambda} \right], \quad \tilde{c}_\ell(\mu) \simeq \tilde{c}_\ell(\Lambda) \left[1 + \frac{\gamma_{\tilde{c}\tilde{c}}}{16\pi^2} \ln \frac{\mu}{\Lambda} \right].$$

These factors cancel in the EDM- g -2 ratio at the leading log.

Appendix D.4. Pipeline Summary

Starting from ε and a reduced Wilson coefficient $\widehat{C}_{e\gamma}^\ell$ at a large scale,

$$C_{e\gamma}^\ell(\Lambda) = \varepsilon \widehat{C}_{e\gamma}^\ell(\Lambda) \implies \begin{cases} \text{SMEFT running: } C_{e\gamma}^\ell(\mu_{\text{EW}}) \text{ via (A12),} \\ \text{Matching: } c_\ell(\mu_{\text{EW}}), \tilde{c}_\ell(\mu_{\text{EW}}), \\ \text{LEFT running: } c_\ell(m_\ell), \tilde{c}_\ell(m_\ell), \\ \text{Observables: } \Delta a_\ell = \frac{2m_\ell}{e} \Re c_\ell, \quad d_\ell = -\tilde{c}_\ell. \end{cases}$$

Appendix E. Phenomenological Tables, Units, and Uncertainty Propagation

Appendix E.1. Masses, Kinematic Factors, and Unit Conversions

We use PDG pole masses (to sufficient accuracy for our purposes):

$$m_e = 0.51099895 \text{ MeV}, \quad m_\mu = 105.6583755 \text{ MeV}, \quad m_\tau = 1776.86 \text{ MeV}.$$

Useful ratios:

$$\frac{m_e}{m_\mu} \approx 4.83633 \times 10^{-3}, \quad \left(\frac{m_e}{m_\mu}\right)^2 \approx 2.339 \times 10^{-5}, \quad \frac{m_\tau}{m_\mu} \approx 16.815.$$

In natural units, $\hbar = c = 1$, $1 \text{ GeV}^{-1} = 1.97327 \times 10^{-14} \text{ cm}$. Hence, for EDMs in $e \cdot \text{cm}$,

$$d_\ell [e \cdot \text{cm}] = \left(\frac{e}{2m_\ell}\right) \Delta a_\ell \tan \phi_\ell \times (1.97327 \times 10^{-14} \text{ cm}).$$

With $e = \sqrt{4\pi\alpha}$, the overall numerical factor is often absorbed into the definition of d_ℓ ; here we keep the compact proportionality form consistent with (63).

Appendix E.2. Texture Predictions in Terms of Δa_μ

Let Δa_μ parametrize calibration and assume reduced coefficient ratios $\rho_\ell = \widehat{C}_{e\gamma}^\ell / \widehat{C}_{e\gamma}^\mu$.

Texture U (universal).

$$\Delta a_e = \Delta a_\mu \left(\frac{m_e}{m_\mu}\right) \approx 4.8363 \times 10^{-3} \Delta a_\mu, \quad \Delta a_\tau = \Delta a_\mu \left(\frac{m_\tau}{m_\mu}\right) \approx 16.815 \Delta a_\mu.$$

EDMs:

$$d_\ell = \frac{e}{2m_\ell} \Delta a_\ell \tan \phi_\ell, \quad \text{so} \quad \frac{d_e}{d_\mu} = 1, \quad \frac{d_\tau}{d_\mu} = 1.$$

Texture Y (Yukawa-aligned).

$$\Delta a_e = \Delta a_\mu \left(\frac{m_e}{m_\mu}\right)^2 \approx 2.339 \times 10^{-5} \Delta a_\mu, \quad \Delta a_\tau = \Delta a_\mu \left(\frac{m_\tau}{m_\mu}\right)^2 \approx 282.8 \Delta a_\mu.$$

EDMs:

$$\frac{d_e}{d_\mu} = \frac{m_e}{m_\mu} \approx 4.8363 \times 10^{-3}, \quad \frac{d_\tau}{d_\mu} = \frac{m_\tau}{m_\mu} \approx 16.815.$$

Appendix E.3. Uncertainty Propagation

Let σ_μ be the uncertainty on Δa_μ and σ_ϕ on ϕ_ℓ . Then

$$\sigma(\Delta a_e) = \left| \frac{\partial \Delta a_e}{\partial \Delta a_\mu} \right| \sigma_\mu = \begin{cases} \frac{m_e}{m_\mu} \sigma_\mu & (\text{Texture U}), \\ \left(\frac{m_e}{m_\mu}\right)^2 \sigma_\mu & (\text{Texture Y}). \end{cases}$$

For EDMs,

$$d_\ell = \left(\frac{e}{2m_\ell}\right) \Delta a_\ell \tan \phi_\ell \Rightarrow \sigma(d_\ell) = \left| \frac{e}{2m_\ell} \right| \sqrt{\tan^2 \phi_\ell \sigma^2(\Delta a_\ell) + \Delta a_\ell^2 \sec^4 \phi_\ell \sigma^2(\phi_\ell)}.$$

Appendix E.4. Illustrative Tables (Symbolic)

Table A1. Parameter-free mass-scaling predictions.

	$\Delta a_e / \Delta a_\mu$	$\Delta a_\tau / \Delta a_\mu$	d_e / d_μ
Texture U	m_e / m_μ	m_τ / m_μ	1
Texture Y	$(m_e / m_\mu)^2$	$(m_\tau / m_\mu)^2$	m_e / m_μ

References

- Adler, S.L. Axial-vector vertex in spinor electrodynamics. *Phys. Rev.* **1969**, *177*, 2426–2438. [\[CrossRef\]](#)
- Bell, J.S.; Jackiw, R. A PCAC puzzle: $\pi^0 \rightarrow \gamma\gamma$ in the σ -model. *Nuovo Cim. A* **1969**, *60*, 47–61. [\[CrossRef\]](#)
- Duff, M.J. Twenty years of the Weyl anomaly. *Class. Quantum Grav.* **1994**, *11*, 1387–1404. [\[CrossRef\]](#)
- Wess, J.; Zumino, B. Consequences of anomalous Ward identities. *Phys. Lett. B* **1971**, *37*, 95–97. [\[CrossRef\]](#)
- Alvarez-Gaumé, L.; Witten, E. Gravitational anomalies. *Nucl. Phys. B* **1984**, *234*, 269–330. [\[CrossRef\]](#)
- Brunetti, R.; Fredenhagen, K. Microlocal analysis and interacting quantum field theories: Renormalization on physical backgrounds. *Commun. Math. Phys.* **2000**, *208*, 623–661. [\[CrossRef\]](#)
- Rejzner, K. *Perturbative Algebraic Quantum Field Theory: An Introduction for Mathematicians*; Mathematical Physics Studies; Springer: Berlin/Heidelberg, Germany, 2016.
- Hollands, S.; Wald, R.M. Local Wick polynomials and time ordered products of quantum fields in curved spacetime. *Commun. Math. Phys.* **2001**, *223*, 289–326. [\[CrossRef\]](#)
- Hollands, S.; Wald, R.M. Existence of local covariant time ordered products of quantum fields in curved spacetime. *Commun. Math. Phys.* **2002**, *231*, 309–345. [\[CrossRef\]](#)
- Stueckelberg, E.C.G.; Petermann, A. La normalisation des constantes dans la théorie des quanta. *Helv. Phys. Acta* **1953**, *26*, 499–520.
- Brunetti, R.; Dütsch, M.; Fredenhagen, K. Perturbative algebraic quantum field theory and the renormalization groups. *Adv. Theor. Math. Phys.* **2009**, *13*, 1541–1599. [\[CrossRef\]](#)
- Gerstenhaber, M. On the deformation of rings and algebras. *Ann. Math.* **1964**, *79*, 59–103. [\[CrossRef\]](#)
- Haag, R. *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed.; Springer: Berlin/Heidelberg, Germany, 1996.
- Brunetti, R.; Fredenhagen, K.; Verch, R. The generally covariant locality principle—A new paradigm for local quantum physics. *Commun. Math. Phys.* **2003**, *237*, 31–68. [\[CrossRef\]](#)
- Hörmander, L. *The Analysis of Linear Partial Differential Operators I*; Springer: Berlin/Heidelberg, Germany, 1983.
- Scharf, G. *Finite Quantum Electrodynamics: The Causal Approach*, 2nd ed.; Springer: Berlin/Heidelberg, Germany, 1995.
- Steinmann, O. *Perturbation Expansions in Axiomatic Field Theory*; Lecture Notes in Physics 11; Springer: Berlin/Heidelberg, Germany, 1971.
- Bär, C.; Fredenhagen, K. (Eds.) *Quantum Field Theory on Curved Spacetimes: Concepts and Mathematical Foundations*; Lecture Notes in Physics 786; Springer: Berlin/Heidelberg, Germany, 2009.
- Radzikowski, M.J. Micro-local approach to the Hadamard condition in quantum field theory on curved space-time. *Commun. Math. Phys.* **1996**, *179*, 529–553. [\[CrossRef\]](#)
- Hochschild, G. On the cohomology groups of an associative algebra. *Ann. Math.* **1945**, *46*, 58–67. [\[CrossRef\]](#)
- Cartan, H.; Eilenberg, S. *Homological Algebra*; Princeton University Press: Princeton, NJ, USA, 1956.
- Gerstenhaber, M. The cohomology structure of an associative ring. *Ann. Math.* **1963**, *78*, 267–288. [\[CrossRef\]](#)
- Nijenhuis, A.; Richardson, R.W., Jr. Cohomology and deformations in algebraic structures. *Bull. Am. Math. Soc.* **1966**, *72*, 1–29. [\[CrossRef\]](#)
- Loday, J.-L.; Vallette, B. *Algebraic Operads*; Grundlehren der Mathematischen Wissenschaften 346; Springer: Berlin/Heidelberg, Germany, 2012.
- Epstein, H.; Glaser, V. The role of locality in perturbation theory. *Ann. Inst. H. Poincaré A* **1973**, *19*, 211–295.
- Bayen, F.; Flato, M.; Fronsdal, C.; Lichnerowicz, A.; Sternheimer, D. Deformation theory and quantization. *Ann. Phys.* **1978**, *111*, 61–151. [\[CrossRef\]](#)
- Fedosov, B.V. A simple geometrical construction of deformation quantization. *J. Diff. Geom.* **1994**, *40*, 213–238. [\[CrossRef\]](#)
- Kontsevich, M. Deformation quantization of Poisson manifolds. *Lett. Math. Phys.* **2003**, *66*, 157–216. [\[CrossRef\]](#)
- Rieffel, M.A. *Deformation Quantization for Actions of $\mathbb{R}^d$* ; American Mathematical Society: Providence, RI, USA, 1993; Volume 106.
- Grosse, H.; Lechner, G. Wedge-local quantum fields and noncommutative Minkowski space. *Commun. Math. Phys.* **2007**, *276*, 491–508. [\[CrossRef\]](#)
- Buchholz, D.; Lechner, G.; Summers, S.J. Warped convolutions, Rieffel deformations and the construction of quantum field theories. *Commun. Math. Phys.* **2011**, *304*, 95–123. [\[CrossRef\]](#)

32. Dappiaggi, C.; Hack, T.-P.; Pinamonti, N. The extended algebra of observables for Dirac fields and the trace anomaly of their stress-energy tensor. *Rev. Math. Phys.* **2009**, *21*, 1241–1312. [[CrossRef](#)]
33. Zahn, J. The renormalized locally covariant Dirac field. *Rev. Math. Phys.* **2014**, *26*, 1330012. [[CrossRef](#)]
34. Hollands, S. The operator product expansion for perturbative quantum field theory in curved spacetime. *Commun. Math. Phys.* **2007**, *273*, 1–36. [[CrossRef](#)]
35. Hollands, S.; Wald, R.M. Quantum field theory in curved spacetime, the operator product expansion, and dark energy. *Gen. Relativ. Gravit.* **2008**, *40*, 2051–2059. [[CrossRef](#)]
36. Ward, J.C. An identity in quantum electrodynamics. *Phys. Rev.* **1950**, *78*, 182. [[CrossRef](#)]
37. Takahashi, Y. On the generalized Ward identity. *Nuovo Cim.* **1957**, *6*, 371–375. [[CrossRef](#)]
38. Piguet, O.; Sorella, S.P. *Algebraic Renormalization: Perturbative Renormalization, Symmetries and Anomalies*; Lecture Notes in Physics Monographs 28; Springer: Berlin/Heidelberg, Germany, 1995.
39. Barnich, G.; Brandt, F.; Henneaux, M. Local BRST cohomology in gauge theories. *Phys. Rep.* **2000**, *338*, 439–569. [[CrossRef](#)]
40. Bardeen, W.A.; Zumino, B. Consistent and covariant anomalies in gauge and gravitational theories. *Nucl. Phys. B* **1984**, *244*, 421–453. [[CrossRef](#)]
41. Mañes, J.; Stora, R.; Zumino, B. Algebraic study of chiral anomalies. *Commun. Math. Phys.* **1985**, *102*, 157–174. [[CrossRef](#)]
42. Sorella, S.P. Algebraic characterization of the Wess–Zumino consistency conditions in gauge theories. *Commun. Math. Phys.* **1993**, *157*, 231–243. [[CrossRef](#)]
43. Deser, S.; Schwimmer, A. Geometric Classification of Conformal Anomalies in Arbitrary Dimensions. *Phys. Lett. B* **1993**, *309*, 279–284. [[CrossRef](#)]
44. Rejzner, K. BV quantization in perturbative algebraic QFT. *arXiv* **2020**, arXiv:2004.14272.
45. Brunetti, R.; Dütsch, M.; Fredenhagen, K.; Rejzner, K. Unitary, anomalous Master Ward Identity and its connections to the Wess–Zumino condition, BV formalism and L_∞ -algebras. *Ann. Henri Poincaré* **2024**, *25*, 2547–2583. [[CrossRef](#)]
46. Fujikawa, K. Path-Integral Measure for Gauge-Invariant Fermion Theories. *Phys. Rev. Lett.* **1979**, *42*, 1195. [[CrossRef](#)]
47. Fujikawa, K. Path integral for gauge theories with fermions. *Phys. Rev. D* **1980**, *21*, 2848. Erratum in: *Phys. Rev. D* **1980**, *22*, 1499. [[CrossRef](#)]
48. Fredenhagen, K.; Rejzner, K. Batalin–Vilkovisky formalism in perturbative algebraic quantum field theory. *Commun. Math. Phys.* **2013**, *317*, 697–725. [[CrossRef](#)]
49. Grzadkowski, B.; Iskrzyński, M.; Misiak, M.; Rosiek, J. Dimension-Six Terms in the Standard Model Lagrangian. *J. High Energy Phys.* **2010**, *10*, 085. [[CrossRef](#)]
50. Jenkins, E.E.; Manohar, A.V.; Trott, M. Renormalization Group Evolution of the Standard Model Dimension Six Operators I: Formalism and λ Dependence. *J. High Energy Phys.* **2013**, *10*, 087. [[CrossRef](#)]
51. Jenkins, E.E.; Manohar, A.V.; Trott, M. Renormalization Group Evolution of the Standard Model Dimension Six Operators II: Yukawa Dependence. *J. High Energy Phys.* **2014**, *1*, 35. [[CrossRef](#)]
52. Alonso, R.; Jenkins, E.E.; Manohar, A.V.; Trott, M. Renormalization Group Evolution of the Standard Model Dimension Six Operators III: Gauge Coupling Dependence and Phenomenology. *J. High Energy Phys.* **2014**, *04*, 159. [[CrossRef](#)]
53. Bennett, G.W.; Muon $g-2$ Collaboration. Final Report of the Muon E821 Anomalous Magnetic Moment Measurement at BNL. *Phys. Rev. D* **2006**, *73*, 072003. [[CrossRef](#)]
54. Abi, B.; Muon $g-2$ Collaboration. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm. *Phys. Rev. Lett.* **2021**, *126*, 141801. [[CrossRef](#)]
55. Aoyama, T.; Asmussen, N.; Benayoun, M.; Bijmns, J.; Blum, T.; Bruno, M.; Caprini, I.; Calame, C.M.C.; Cè, M.; Colangelo, G.; et al. The anomalous magnetic moment of the muon in the Standard Model. *Phys. Rep.* **2020**, *887*, 1–166. [[CrossRef](#)]
56. Jegerlehner, F.; Nyffeler, A. The Muon $g-2$. *Phys. Rep.* **2009**, *477*, 1–110. [[CrossRef](#)]
57. Hanneke, D.; Fogwell, S.; Gabrielse, G. New Measurement of the Electron Magnetic Moment and the Fine Structure Constant. *Phys. Rev. Lett.* **2008**, *100*, 120801. [[CrossRef](#)]
58. Morel, L.; Yao, Z.; Cladé, P.; Guellati-Khélifa, S. Determination of the fine-structure constant with an accuracy of 81 parts per trillion. *Nature* **2020**, *588*, 61–65. [[CrossRef](#)]
59. Drinfel'd, V.G. Quantum groups. In Proceedings of the International Congress of Mathematicians, Berkeley, CA, USA, 3–11 August 1986; Volume 1, pp. 798–820.
60. Majid, S. *Foundations of Quantum Group Theory*; Cambridge University Press: Cambridge, MA, USA, 1995.
61. Aschieri, P.; Blohmann, C.; Dimitrijević, M.; Meyer, F.; Schupp, P.; Wess, J. A gravity theory on noncommutative spaces. *Class. Quantum Grav.* **2005**, *22*, 3511–3532. [[CrossRef](#)]
62. Chaichian, M.; Kulish, P.P.; Nishijima, K.; Tureanu, A. On a Lorentz-invariant interpretation of noncommutative space-time and its implications on noncommutative QFT. *Phys. Lett. B* **2004**, *604*, 98–102. [[CrossRef](#)]
63. Lechner, G. Construction of quantum field theories with factorizing S -matrices. *Commun. Math. Phys.* **2008**, *277*, 821–860. [[CrossRef](#)]

64. Markl, M.; Shnider, S.; Stasheff, J. *Operads in Algebra, Topology and Physics*; Mathematical Surveys and Monographs 96; American Mathematical Society (AMS): Providence, RI, USA, 2002.
65. Getzler, E. Lie theory for nilpotent L_∞ -algebras. *Ann. Math.* **2009**, *170*, 271–301. [[CrossRef](#)]
66. Pirashvili, T. Hodge decomposition for higher order Hochschild homology. *Ann. Sci. Éc. Norm. Supér.* **2000**, *33*, 151–179. [[CrossRef](#)]
67. Ginot, G. Higher order Hochschild cohomology. *C. R. Math. Acad. Sci. Paris* **2008**, *346*, 625–628. [[CrossRef](#)]
68. Ayala, D.; Francis, J. Factorization homology of topological manifolds. *J. Topol.* **2015**, *8*, 1045–1084. [[CrossRef](#)]
69. Costello, K.; Gwilliam, O. *Factorization Algebras in Quantum Field Theory*; Cambridge University Press: Cambridge, MA, USA, 2016; Volume 1.
70. Connes, A.; Marcolli, M. *Noncommutative Geometry, Quantum Fields and Motives*; AMS Colloquium Publications 55; American Mathematical Society (AMS): Providence, RI, USA, 2007.
71. Ebrahimi-Fard, K.; Guo, L.; Kreimer, D. Spitzer's identity and the algebraic Birkhoff decomposition in p QFT. *J. Phys. A* **2004**, *37*, 11037–11052. [[CrossRef](#)]
72. Atiyah, M.F.; Singer, I.M. The index of elliptic operators I. *Ann. Math.* **1968**, *87*, 484–530. [[CrossRef](#)]
73. Berline, N.; Getzler, E.; Vergne, M. *Heat Kernels and Dirac Operators*; Springer: Berlin/Heidelberg, Germany, 1992.
74. Callan, C.G., Jr; Harvey, J.A. Anomalies and fermion zero modes on strings and domain walls. *Nucl. Phys. B* **1985**, *250*, 427–436. [[CrossRef](#)]
75. Zumino, B. Chiral anomalies and differential geometry. In *Relativity, Groups and Topology II*; Les Houches 1983; DeWitt, B.S., Stora, R., Eds.; North-Holland: Amsterdam, The Netherlands, 1984.
76. Freed, D.S. Anomalies and invertible field theories. *Proc. Symp. Pure Math.* **2014**, *88*, 25–45.
77. Freed, D.S.; Hopkins, M.J. Reflection positivity and invertible topological phases. *Geom. Topol.* **2021**, *25*, 1165–1330. [[CrossRef](#)]
78. Witten, E. Global gravitational anomalies. *Commun. Math. Phys.* **1985**, *100*, 197–229. [[CrossRef](#)]
79. Uhlmann, A. Parallel transport for density matrices. *Rep. Math. Phys.* **1986**, *24*, 229–240. [[CrossRef](#)]
80. Petz, D. Monotone metrics on matrix spaces. *Linear Algebra Appl.* **1996**, *244*, 81–96. [[CrossRef](#)]
81. Araki, H. Relative entropy of states of von Neumann algebras. *Publ. RIMS Kyoto* **1976**, *11*, 809–833. [[CrossRef](#)]
82. Bratteli, O.; Robinson, D.W. *Operator Algebras and Quantum Statistical Mechanics II*, 2nd ed.; Springer: Berlin/Heidelberg, Germany, 1997.
83. Zamolodchikov, A.B. Irreversibility of the flux of the renormalization group in a 2D field theory. *JETP Lett.* **1986**, *43*, 730–732.
84. Beilinson, A.; Drinfeld, V. *Chiral Algebras*; AMS Colloquium Publications 51; American Mathematical Society (AMS): Providence, RI, USA, 2004.
85. Lurie, J. On the Classification of Topological Field Theories. *Curr. Dev. Math.* **2008**, *2008*, 129–280. [[CrossRef](#)]
86. Bogoliubov, N.N.; Parasiuk, O.S. On the multiplication of the causal function in the quantum theory of fields. *Acta Math.* **1957**, *97*, 227–266.
87. Hepp, K. Proof of the Bogoliubov-Parasiuk theorem on renormalization. *Commun. Math. Phys.* **1966**, *2*, 301–326. [[CrossRef](#)]
88. Zimmermann, W. Convergence of Bogoliubov's method of renormalization in momentum space. *Commun. Math. Phys.* **1969**, *15*, 208–234. [[CrossRef](#)]
89. Connes, A.; Kreimer, D. Hopf algebras, renormalization and noncommutative geometry. *Commun. Math. Phys.* **1998**, *199*, 203–242. [[CrossRef](#)]
90. Connes, A.; Kreimer, D. Renormalization in quantum field theory and the Riemann–Hilbert problem I: The Hopf algebra structure of graphs and the main theorem. *Commun. Math. Phys.* **2000**, *210*, 249–273. [[CrossRef](#)]
91. Bogoliubov, N.N.; Shirkov, D.V. *Introduction to the Theory of Quantized Fields*; Wiley-Interscience: Hoboken, NJ, USA, 1959.
92. Peierls, R. The commutation laws of relativistic field theory. *Proc. R. Soc. Lond. A* **1952**, *214*, 143–157. [[CrossRef](#)]
93. Peskin, M.E.; Schroeder, D.V. *An Introduction to Quantum Field Theory*; Addison-Wesley: Boston, MA, USA, 1995.
94. D'Ambrosio, G.; Giudice, G.F.; Isidori, G.; Strumia, A. Minimal Flavor Violation: An Effective Field Theory Approach. *Nucl. Phys. B* **2002**, *645*, 155–187. [[CrossRef](#)]
95. Pospelov, M.; Ritz, A. Electric dipole moments as probes of new physics. *Ann. Phys.* **2005**, *318*, 119–169. [[CrossRef](#)]
96. Engel, J.; Ramsey-Musolf, M.J.; van Kolck, U. Electric Dipole Moments of Nucleons, Nuclei, and Atoms: The Standard Model and Beyond. *Prog. Part. Nucl. Phys.* **2013**, *71*, 21–74. [[CrossRef](#)]
97. Chupp, T.; Fierlinger, P.; Ramsey-Musolf, M.; Singh, J. Electric Dipole Moments of Atoms, Molecules, Nuclei, and Particles. *Rev. Mod. Phys.* **2019**, *91*, 015001. [[CrossRef](#)]
98. Gerstenhaber, M. On the deformation of rings and algebras II. *Ann. Math.* **1966**, *84*, 1–19. [[CrossRef](#)]
99. Gerstenhaber, M. On the deformation of rings and algebras III. *Ann. Math.* **1968**, *88*, 1–34. [[CrossRef](#)]
100. Gerstenhaber, M. On the deformation of rings and algebras IV. *Ann. Math.* **1974**, *99*, 257–276. [[CrossRef](#)]
101. Gerstenhaber, M. On the deformation of rings and algebras V. *Ann. Math.* **1974**, *99*, 277–301. [[CrossRef](#)]
102. Yekutieli, A. Deformation Quantization in Algebraic Geometry. *Adv. Math.* **2005**, *198*, 383–432. [[CrossRef](#)]

103. Kontsevich, M. Deformation quantization of Poisson manifolds, I. In *ICM 1998 Proceedings*; Birkhäuser: Basel, Switzerland, 1998.
104. Fredenhagen, K.; Rejzner, K. Perturbative algebraic quantum field theory. In *Mathematical Aspects of Quantum Field Theories*; Springer: Berlin/Heidelberg, Germany, 2015.
105. Brivio, I.; Trott, M. The Standard Model as an Effective Field Theory. *Phys. Rep.* **2019**, *793*, 1–98. [[CrossRef](#)]
106. Aebischer, J.; Brivio, I.; Celis, A.; Evans, J.A.; Jiang, Y.; Kumar, J.; Pan, X.; Porod, W.; Rosiek, J.; Shih, D.; et al. WCxf: An exchange format for Wilson coefficients beyond the Standard Model. *Comput. Phys. Commun.* **2018**, *232*, 71–83. [[CrossRef](#)]
107. Manohar, A.V.; Wise, M.B. *Heavy Quark Physics*; Cambridge Monographs on Particle Physics; Cambridge University Press: Cambridge, MA, USA, 2000.
108. Particle Data Group. Review of Particle Physics. *Prog. Theor. Exp. Phys.* **2024**, *2024*, 083C01.
109. Marciano, W.J.; Masiero, A.; Paradisi, P.; Passera, M. Contributions to a_μ in and beyond the Standard Model. In *Lepton Dipole Moments*; World Scientific: Singapore, 2010.
110. Hörmander, L. *The Analysis of Linear Partial Differential Operators II*; Springer: Berlin/Heidelberg, Germany, 1983.
111. Duistermaat, J.J. *Fourier Integral Operators*; Birkhäuser: Basel, Switzerland, 1996.
112. Khavkine, I.; Moretti, V. Algebraic QFT in curved spacetime and microlocal analysis. In *Advances in Algebraic Quantum Field Theory*; Springer: Berlin/Heidelberg, Germany, 2015.
113. Hollands, S.; Wald, R.M. Quantum field theory in curved spacetime, the operator product expansion, and perturbative quantum gravity. *Commun. Math. Phys.* **2010**, *293*, 85–125. [[CrossRef](#)]
114. Rejzner, K. *Perturbative Algebraic Quantum Field Theory*; Springer Briefs in Mathematical Physics; Springer: Berlin/Heidelberg, Germany, 2016.

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